

PET2Rep: Towards Vision-Language Model-Driven Automated Radiology Report Generation for Positron Emission Tomography

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Abstract

Positron emission tomography (PET) is a cornerstone of modern oncologic and neurologic imaging, distinguished by its unique ability to illuminate dynamic metabolic processes that transcend the anatomical focus of traditional imaging technologies. Radiology reports are essential for clinical decision making, yet their manual creation is labor-intensive and time-consuming. Recent advancements of vision-language models (VLMs) have shown strong potential in medical applications, presenting a promising avenue for automating report generation. However, existing applications of VLMs in the medical domain have predominantly focused on structural imaging modalities, while the unique characteristics of molecular PET imaging have largely been overlooked. To bridge the gap, we introduce PET2Rep, a large-scale comprehensive benchmark for evaluation of general and medical VLMs for radiology report generation for PET images. PET2Rep stands out as the first dedicated dataset for PET report generation with metabolic information, uniquely capturing whole-body image-report pairs that cover dozens of organs to fill the critical gap in existing benchmarks and mirror real-world clinical comprehensiveness. In addition to widely recognized natural language generation metrics, we introduce a series of clinical efficacy metrics to evaluate the quality of radiotracer uptake pattern description in key organs in generated reports. We conduct a head-to-head comparison of 30 cutting-edge general-purpose and medical-specialized VLMs. The results show that the current state-of-the-art VLMs perform poorly on PET report generation task, falling considerably short of fulfilling practical needs. Moreover, we identify several key insufficiency that need to be addressed to advance the development in medical applications. We believe PET2Rep will serve as a platform for the development and application of VLMs for PET imaging, accelerating the development of trustworthy reporting tools that can genuinely alleviate radiologist burden and enhance patient care.

Project page: <https://github.com/YichiZhang98/PET2Rep>.

Introduction

Radiology plays a crucial role in modern healthcare, enabling clinicians to visualize internal structures of patients and make informed decisions for diagnosis and treatment. Positron Emission Tomography (PET) stands as a corner-

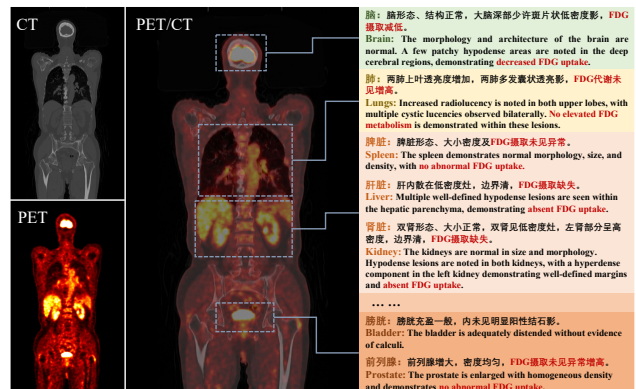


Figure 1: An overview of PET2Rep benchmark. Each case contains whole-body PET/CT images with radiology report.

stone in contemporary oncological and neurological imaging, offering unparalleled insights into dynamic metabolic processes. Unlike imaging modalities like X-ray and CT which primarily focus on information of anatomical structures, PET excels at visualizing metabolic information of physiological functions. By tracking the distribution of radioactive tracers, PET can detect early signs of disease progression, monitor treatment response, and guide personalized therapy plans (Peng et al. 2023; Xue et al. 2024). This functional imaging capability has revolutionized the diagnosis and management of various conditions (Schwenck et al. 2023). In the clinical workflow, radiology reports play a pivotal role in translating imaging into actionable information for healthcare providers. These reports summarize the radiologist’s interpretation of the images, highlight key observations, and provide recommendations for further evaluation or treatment (Pang, Li, and Zhao 2023). However, the process of manually creating these reports is inherently labor-intensive and time-consuming, often burdening radiologists with a significant administrative workload. This not only limits their capacity to handle a growing volume of imaging studies, but also introduces potential delays in patient care (Ashraf et al. 2023).

The recent surge in vision-language models (VLMs) has attracted interest from the medical community, where these

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models hold great potential to transform various aspects of clinical practice like automatic medical report generation (Zhang et al. 2024). Leveraging the power of large-scale pre-training, VLMs can analyze medical images and generate corresponding textual descriptions, effectively bridging the gap between visual data and clinical language. However, existing applications of VLMs in the medical domain have predominantly focused on structural imaging modalities (Liu et al. 2024; Hamamci et al. 2024; Zhu et al. 2025b), while the unique characteristics and clinical value of PET imaging have largely been overlooked in the current research landscape. As the analysis of PET images poses unique challenges due to the need to integrate functional and anatomical information and specialized knowledge required to interpret tracer uptake patterns (Coleman et al. 2010; Matsubara et al. 2022), it is worth rethinking that *How Far are VLMs from Effective Radiology Report Generation for Positron Emission Tomography Imaging?*

To answer this question, we introduce PET2Rep, a comprehensive benchmark for the evaluation of radiology report generation for PET imaging. Compared with existing medical benchmarks, the key advantages of PET2Rep can be concluded in the following three aspects.

The First PET/CT Report Dataset. PET2Rep is the first dataset dedicated to PET/CT report generation. Unlike other modalities like X-ray and CT which primarily focus on anatomical structures, PET operates at the molecular level, enabling the assessment of metabolic information. This unique feature allows for early disease identification, often before anatomical changes are visible on other imaging modalities (Gatidis et al. 2024). PET2Rep is a large-scale multi-modal dataset of 565 cases with paired PET, CT and corresponding radiology reports. Given the high cost of PET/CT scans and the need for specialized expertise in report writing, there is currently no relevant dataset available, which highlights the importance of PET2Rep in advancing research in this field.

Whole-Body Imaging with Radiology Reports. Existing medical imaging benchmarks are often limited to specific anatomical domains. For instance, chest X-ray report generation primarily address thoracic pathologies (Liu et al. 2024), while those for CT reports concentrate on the volume and morphology of organs and lesions in the chest (Hamamci et al. 2024) or abdominal regions (Bassi et al. 2025). In contrast, PET2Rep encompasses a much broader anatomical scope, with images ranging from the head and neck to the proximal limbs. Consequently, its corresponding reports provide detailed evaluations of dozens of organs body-wide, demanding a more extensive scope of medical knowledge for accurate interpretation, as shown in Figure.1. This holistic approach more closely simulates real-world oncology practice, where radiologists conduct comprehensive assessments rather than focusing on isolated areas.

Data Collection from Clinical Scenarios. Many existing medical multimodal benchmarks are developed from public imaging archives (Sepehri et al. 2024; Chen et al. 2024). These frameworks often generate tasks that probe for superficial understanding of the image, such as identifying the imaging modality or naming marked organs, rather than

complex clinical reasoning (Ye et al. 2024; Zhou et al. 2025). Such scenarios test for basic medical knowledge and differ significantly from the complex demands of a real clinical workflow. In contrast, PET2Rep is collected from real clinical scenarios and incorporates data directly from the clinical setting, ensuring that the benchmark authentically reflects the challenges radiologists encounter in their daily work. This ensures the authenticity and clinical relevance of the PET2Rep benchmark while minimizing the risk of data leakage, thereby reflecting the generalization performance of VLMs in real-world clinical scenarios.

To make a comprehensive evaluation of the performance of VLMs, we establish a standardized evaluation pipeline for PET/CT radiology report generation. We formulate a prompting framework incorporating essential elements including imaging modality specifications and clinical objectives and design a structured report template aligned with radiological training protocols. This approach ensures faithful translation of image-derived information into formatted reports that maintain consistency with expert-generated radiological reports. We conduct a comprehensive evaluation state-of-the-art models, including 19 general purpose and 11 medical-specific VLMs on PET2Rep benchmark. The experimental results show that current cutting-edge VLMs exhibit suboptimal performance on the task, falling considerably short of fulfilling real-world requirements. Furthermore, our analysis reveals several critical limitations that must be tackled to drive progress in clinical applications.

Related Works

Positron Emission Tomography

Positron Emission Tomography (PET) is a clinical imaging technique that reveals ongoing metabolic processes in the body by detecting gamma photons generated from positron annihilation after injecting radioactive tracers. As the most widely used tracer, fluorodeoxyglucose (FDG) assesses local glucose uptake to evaluate organ metabolism and detect tumor metastasis, enabling monitoring of treatment progress (Ren et al. 2019). Clinically, PET is primarily used for early tumor screening for cancer detection (Gatidis et al. 2024; Peng et al. 2023), organ metabolic function assessment (Xue et al. 2024; Zhang et al. 2025), and treatment monitoring (van der Geest et al. 2021). The unique metabolic imaging capability provides indispensable insights for disease diagnosis and treatment optimization in clinical applications.

Vision-Language Models

Vision-Language Models (VLMs) have emerged as a transformative development in artificial intelligence, effectively bridging the gap between visual perception and natural language understanding (Zhang et al. 2024). The swift progress in VLM development is largely attributed to innovative pre-training strategies and architectural designs, which have demonstrated remarkable capabilities across a wide array of tasks such as visual question answering and image captioning (Chen et al. 2022; Feng et al. 2025; Lin et al. 2025). Beyond general-purpose vision tasks, VLMs are making significant inroads into specialized fields like medical image

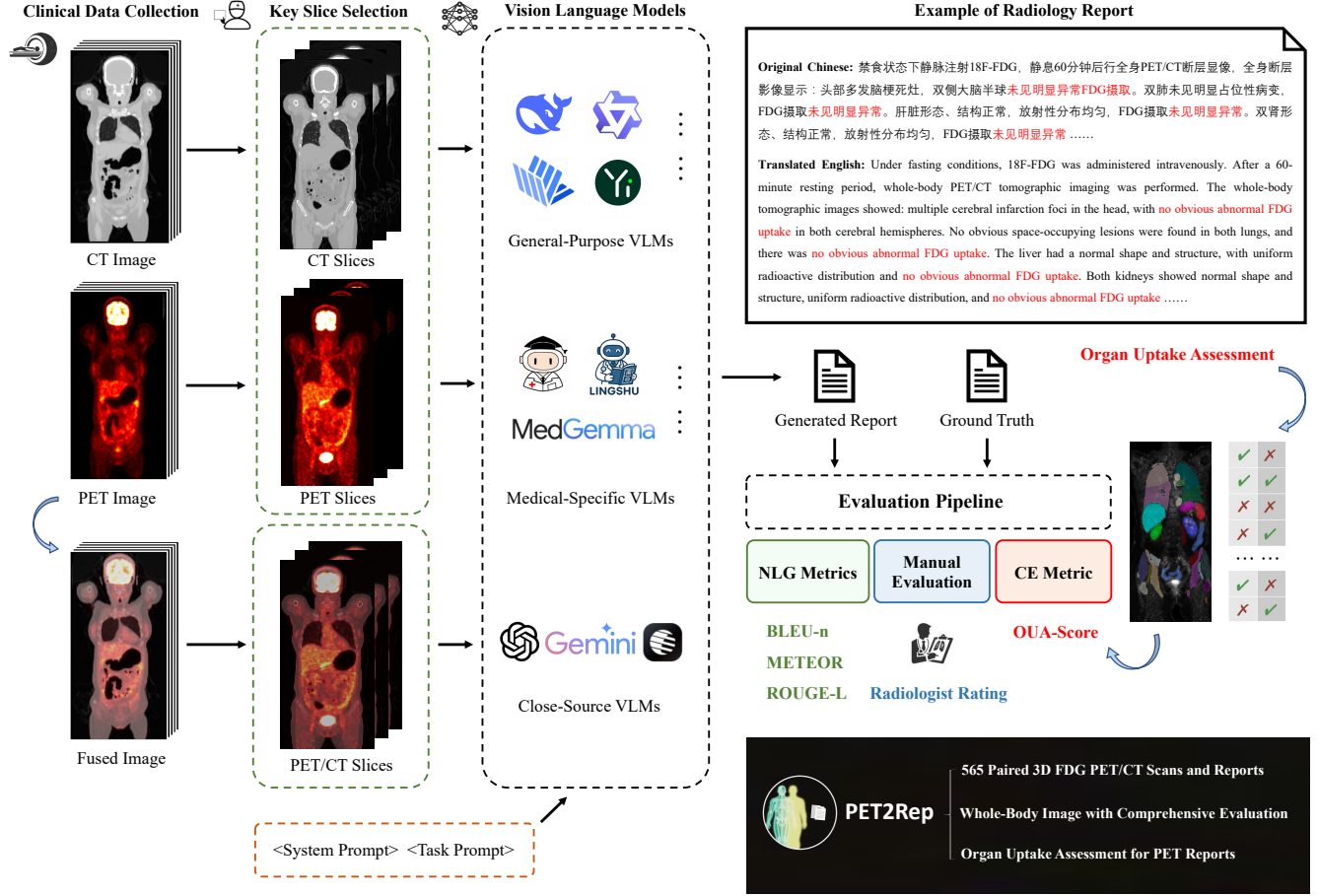


Figure 2: Pipeline of the PET2Rep benchmark for evaluation of VLM-based PET radiology report generation. First, PET/CT images are analyzed by VLMs with a designed prompt format to include necessary information such as image modality, clinical task, and designed report template based on radiologist training guidelines. Then the generated reports are evaluated against the ground-truth reports with widely recognized natural language generation (NLG) metrics and a novel clinical efficacy (CE) metric for PET imaging. We further conduct manual scoring by radiologists for more comprehensive evaluation.

analysis (Peng et al. 2025). VLMs can generate diagnostic reports, answer clinical questions, and highlight regions of interest, offering substantial support to healthcare professionals and promising to enhance the efficacy and accuracy of medical diagnoses (Zhang, Shen, and Jiao 2024; Jiang et al. 2024; Lin, Xu, and Qin 2025).

Radiology Report Generation

The core task of radiology report generation is to transform medical imaging information into accurate and standardized textual reports. Early studies primarily focused on training encoder-decoder architectures for report generation, where the image features are extracted by an encoder and then fed into a decoder to predict the corresponding report (Jing, Xie, and Xing 2018). Given the complexity and the inherent variability in radiological findings, several approaches utilize

confidential guidance or attention mechanism to enhance the adaptability (Song et al. 2022; Wang et al. 2024). Due to the impressive performance in a variety of downstream tasks (Zhang et al. 2024), there has been a surge of investigating VLMs for radiology report generation (Hamamci, Er, and Menze 2024; Chen et al. 2025).

PET2Rep Benchmark

We introduce PET2Rep, a comprehensive benchmark designed to evaluate the performance of VLMs for generating radiology reports from PET images. PET2Rep is the first PET/CT dataset with paired structured radiology reports. A key distinction of PET2Rep is its data sourcing. Unlike benchmarks that rely on data aggregated from online repositories, all data in our work were meticulously collected from actual clinical scenarios. This approach guarantees the au-

thenticity and clinical relevance of the benchmark. Furthermore, by sourcing directly from clinical settings, we mitigate the risk of data leakage, ensuring that PET2Rep accurately reflects the complexity and diversity of real-world radiological practice. The comprehensive workflow of the PET2Rep benchmark is detailed in Figure. 2 and will be elaborated upon in the subsequent sections.

Dataset Construction

We collect 565 cases of 3D whole-body FDG PET/CT imaging from one local medical center, which is the most widely used PET tracer in oncology. As a non-specific tracer, FDG can be used for whole-body imaging to reflect tissue glucose metabolism, which makes the imaging useful in assessing the systemic distribution and metastasis of tumors. Structured radiology reports are constructed based on radiologist-designed templates, which play a pivotal role in assisting physicians to interpret whole-body PET/CT scans in a standardized and organized manner, thereby enhancing clarity and supporting clinical decision-making. The report provides a detailed and objective description of the findings from the PET/CT images in a systematic, head-to-toe sequence, ensuring that no anatomical region is overlooked. It enumerates all detected abnormalities without offering interpretative conclusions, serving as a factual foundation for subsequent clinical assessment. More detailed information and examples of the dataset are shown in the Appendix.

Data Preprocessing

To ensure the accuracy and reliability of multi-modal image analysis, rigorous data pre-processing is indispensable for bringing all imaging modalities into a consistent and interpretable format. In PET/CT imaging, a critical pre-processing step involves resampling the CT images to match the lower spatial resolution of the PET images. This coregistration process aligns both modalities to a common matrix size, ensuring voxel-wise correspondence across datasets. Following resampling, the CT intensities are standardized using z-score normalization to reduce inter-scan variability. Additionally, normalization of PET data is performed by converting the raw radioactivity counts into Standardized Uptake Values (SUV) a widely adopted quantitative metric in PET imaging that accounts for factors such as the injected radiotracer dose and the patient’s body weight (Lucignani, Paganelli, and Bombardieri 2004). To emulate the clinical diagnostic workflow, we implement a fusion process that integrates PET and CT scans. This approach combines the functional information from PET with the anatomical detail provided by CT, reflecting the manner in which radiologists interpret these modalities in clinical practice. The resulting composite image enables visualization of metabolic activity within its precise anatomical context. Such integration is essential for accurately localizing regions of abnormal radiotracer uptake and facilitating a comprehensive assessment of the patient’s condition.

Key Slice Selection

Given that the original PET/CT images are three-dimensional, while most existing VLMs are designed for

2D images, it becomes necessary to select out representative 2D slices from the 3D imaging. In this study, we select the coronal plane as the view for slice sampling, following clinical conventions in which radiologists commonly utilize this view for comprehensive head-to-toe assessments. The coronal plane offers an optimal perspective, capturing the global anatomical context and encompassing the majority of key organs. By analyzing multiple coronal slices, VLMs can effectively capture the salient information embedded within the full 3D scan. Building upon this design, we further emulate the diagnostic process of radiologists and design two strategies for report generation as described below.

Input Separate PET and CT Images. In this strategy, we maintain the distinction between functional and anatomical information by providing the model with two distinct, parallel inputs. For each anatomical location of interest, we extract a corresponding pair of 2D slices: a grayscale slice from the CT volume for structural context, and a pseudo-colored slice from the PET volume to highlight metabolic activity. Specifically, we identify three key locations for analysis, resulting in a total input of six images for the VLMs (a PET/CT pair for each location). This dual-input approach compels the model to learn the complex correlations between anatomical structure and functional uptake, mirroring the cognitive process of a radiologist integrating two different sets of images.

Input Fused PET/CT Images. This strategy involves pre-integrating the multimodal information into a single image before presenting it to the model. For each selected location, we generate a fused image by superimposing the pseudo-colored PET slice directly onto its corresponding grayscale CT slice. In our implementation, we create these fused images for three key locations, providing the model with a total of three input PET/CT images. Each fused image presents an immediate composite view, in which metabolic hotspots are precisely localized within their anatomical context. This method simulates the final visualization that radiologists use for diagnosis. By supplying the model with pre-fused inputs, we eliminate the need for it to learn the fusion process, enabling it to focus directly on understanding the integrated functional and structural patterns.

Experimental Setup

In this study, we evaluate a range of VLMs encompassing both medical-specific and general-purpose models including open-source options and those accessible via proprietary APIs. The weights of open-source models were sourced from respective official Hugging Face repositories. To guide the models in generating radiology-style reports, we design a standardized prompt format specifying the imaging modality, clinical task, and a report template derived from radiologist training guidelines. This ensures that image interpretations are expressed in a format consistent with manually authored radiological reports. Our evaluation are conducted in a zero-shot setting, which serves as a stringent test of the models’ generalization ability, revealing how well they can handle complex medical imaging tasks without any task-specific fine-tuning. All tests were executed using NVIDIA A100 GPUs with 80GB of memory.

Evaluated Models

To comprehensively assess the performance of VLMs on the PET2Rep benchmark, we conducted a systematic evaluation of diverse state-of-the-art VLMs, spanning general-purpose models, medical-specific models, and closed-source models tested via API access.

General-Purpose VLMs. General-purpose VLMs are trained on large and diverse datasets to perform a wide spectrum of multimodal tasks. Their broad training enables strong visual understanding and reasoning capabilities, with versatility and scalability. We select following general-purpose VLMs for evaluation, including Qwen2.5-VL series (Bai et al. 2025), InternVL3 series (Zhu et al. 2025a), Yi-VL series (Young et al. 2024), LLaVA-V1.5 (Liu et al. 2023a), OmniLMM (Yu et al. 2024), VisualGLM (Du et al. 2022) and Deepseek-VL2 (Wu et al. 2024).

Medical-Specific VLMs. In contrast to general-purpose models, medical-specific VLMs are tailored for clinical applications, emphasizing domain adaptation and integration of specialized medical knowledge. Trained on curated medical datasets and aligned with diagnostic workflows, these models prioritize accuracy and reliability in healthcare settings. We select following medical-specific VLMs for evaluation, including LLaVA-Med (Li et al. 2023), MedFlamingo (Moor et al. 2023), Qilin-Med-VL (Liu et al. 2023b), RadFM (Wu et al. 2023), MedDr (He et al. 2024), HuatuoGPT-Vision (Chen et al. 2024), MedVLM-R1 (Pan et al. 2025), and latest MedGemma series (Sellergren et al. 2025) and Lingshu series (Xu et al. 2025).

Closed-Source VLMs. Closed-source VLMs are developed and maintained by enterprises with inaccessible source code, typically provided to users via APIs for integration into applications. We select following closed-source VLMs for evaluation, including Gemini 2.5 Pro (Comanici et al. 2025), GPT-4o (Hurst et al. 2024), Moonshot-v1 (MoonshotAI 2025) and Qwen-VL-Max (Bai et al. 2023).

Evaluation Metrics

To assess the performance of VLMs in radiology report generation, we compare the generated reports against the ground-truth reports using the following aspects.

Natural Language Generation (NLG) Metrics. In line with existing research, we adopt widely recognized NLG metrics, including BLEU-n (Papineni et al. 2002), METEOR (Banerjee and Lavie 2005), and ROUGE-L (Lin 2004). Specifically, BLEU-n evaluates n-gram overlap between generated and reference reports, ROUGE-L measures alignment via the longest common subsequences, and METEOR accounts for synonyms and paraphrases to capture semantic similarity.

PET Clinical Efficacy (CE) Metrics. NLG metrics primarily focus on word and sentence similarity while neglecting diagnostic accuracy. Reports with opposite diagnostic conclusions may achieve similar NLG scores. Conversely, results with correct uptake assessments but inconsistent formatting in the report text might receive lower NLG scores. Existing studies have explored the proposal of clinical efficacy metrics by utilizing text classifiers to extract abnormality labels for CT report evaluation (Hamamci et al. 2024).

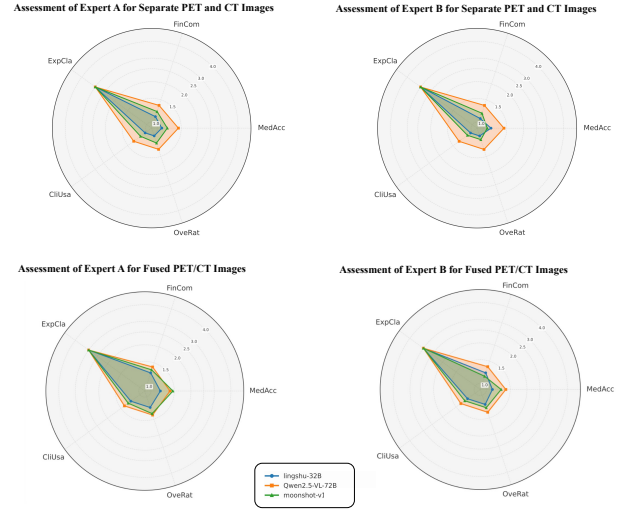


Figure 3: Performance comparison of three VLMs under different task settings for manual evaluation by two radiologists rated across five dimensions, including Medical Accuracy (MedAcc), Key Findings Completeness (FinCom), Expression Clarity (ExpCla), Clinical Usability (CliUsa) and Overall Rating (OverRat).

However, these methods are not applicable to PET reports. To assess the clinical efficacy of PET reports, we introduce a series of CE metrics to evaluate descriptions regarding radiotracer uptake patterns in key organs within generated PET reports. Given that the whole-body PET imaging data used in our study, we extract the assessment of uptake levels corresponding to each key organ from the report text and compare these assessments with the corresponding entries in the ground truth reports. For each key organ, we define four states of radiotracer uptake: *Increased Uptake*, *Decreased Uptake*, *Absent Uptake*, and *Normal*. Given the clinical focus on anomaly detection, we categorize the first three states into three distinct positive classes, with *Normal* serving as the negative class. Our evaluation method involves independently calculating the precision, recall, and F1-score for each of the three positive classes. The final CE metrics are the macro-average across these three positive classes. The implementation details are elaborated in the Appendix. Compared to NLG metrics, CE metrics shift the evaluation from text-matching problem to multi-label classification assessment that more closely aligns with clinical diagnosis.

Results and Analysis

Table. 1 summarizes the performance of the evaluated VLMs, with the last column providing an overall reference score as the average of all metrics. After reviewing the evaluation results, we have drawn following conclusions.

Overall Ineffectiveness or Even Failure. All evaluated VLMs exhibit limited effectiveness in generating structured radiology reports. Alarmingly, most models fail to surpass even a simple template baseline. The requirement to produce comprehensive, whole-body structured reports presents

Model (year/month)	NLG Metrics			CE Metrics			Overall (%)
	BL-4	MTR	RG-L	Pre	Rec	F1	
Template Baseline	0.3150(0.0482)	0.1475(0.0141)	0.5110(0.0319)	0.2282(0.0179)	0.2220(0.0106)	0.2249(0.0123)	27.5
General-Purpose VLMs							
Qwen2.5-VL-7B (25/1)	0.3050(0.0476)	0.1407(0.0198)	0.5075(0.0340)	0.2233(0.0236)	0.1974(0.0083)	0.2094(0.0132)	26.4
	0.3057(0.0467)	0.1390(0.0186)	0.5088(0.0320)	0.2284(0.0227)	0.2023(0.0075)	0.2144(0.0121)	26.6
Qwen2.5-VL-32B (25/1)	0.1777(0.0421)	0.0063(0.0110)	0.4165(0.0516)	0.3402(0.0781)	0.0418(0.0127)	0.0743(0.0214)	17.6
	0.1851(0.0408)	0.0063(0.0111)	0.4295(0.0486)	0.2728(0.0447)	0.0308(0.0047)	0.0554(0.0082)	16.3
Qwen2.5-VL-72B (25/1)	0.2223(0.0585)	0.0655(0.0172)	0.4234(0.0588)	0.2474(0.0513)	0.0295(0.0024)	0.0527(0.0043)	17.3
	0.2273(0.0584)	0.0645(0.0171)	0.4306(0.0594)	0.2917(0.0328)	0.0393(0.0049)	0.0693(0.0084)	18.7
InternVL3-8B (25/4)	0.2439(0.0627)	0.0606(0.0443)	0.4739(0.0630)	0.2425(0.0151)	0.2107(0.0114)	0.2254(0.0119)	24.3
	0.2509(0.0529)	0.0641(0.0463)	0.4845(0.0566)	0.2333(0.0153)	0.2099(0.0074)	0.2208(0.0087)	24.4
InternVL3-14B (25/4)	0.2513(0.0684)	0.0472(0.0528)	0.4835(0.0910)	0.2366(0.0206)	0.2057(0.0095)	0.2199(0.0129)	24.1
	0.2495(0.0671)	0.0532(0.0506)	0.4813(0.0904)	0.2322(0.0196)	0.1982(0.0099)	0.2137(0.0131)	23.8
InternVL3-38B (25/4)	0.1377(0.0924)	0.0775(0.0483)	0.4371(0.1199)	0.2711(0.0203)	0.2072(0.0141)	0.2344(0.0127)	22.8
	0.1446(0.0855)	0.0825(0.0480)	0.4618(0.0913)	0.2674(0.0258)	0.2435(0.0298)	0.2546(0.0278)	24.2
InternVL3-78B (25/4)	0.3090(0.0525)	0.1233(0.0359)	0.4997(0.0401)	0.2355(0.0255)	0.0520(0.0119)	0.0850(0.0157)	21.7
	0.3090(0.0518)	0.1262(0.0318)	0.5008(0.0397)	0.2369(0.0492)	0.0748(0.0083)	0.1132(0.0122)	22.7
Yi-VL-6B (24/1)	0.0065(0.0316)	0.0002(0.0056)	0.0479(0.0709)	0.1144(0.0430)	0.0061(0.0020)	0.0115(0.0038)	3.1
	0.0374(0.0733)	0.0029(0.0165)	0.1156(0.1432)	0.1519(0.0261)	0.0260(0.0033)	0.0444(0.0055)	6.3
Yi-VL-34B (24/1)	0.2610(0.1071)	0.0848(0.0664)	0.4439(0.1420)	0.2305(0.0159)	0.1869(0.0079)	0.2063(0.0098)	23.6
	0.2854(0.0809)	0.0898(0.0645)	0.4779(0.0950)	0.2303(0.0211)	0.2038(0.0072)	0.2160(0.0116)	25.1
LLaVa-V1.5-7B (23/9)	0.1198(0.0508)	0.0126(0.0515)	0.3043(0.0639)	0.2044(0.0287)	0.1022(0.0091)	0.1306(0.0121)	14.6
	0.0328(0.0141)	0.0056(0.0369)	0.1717(0.0283)	0.2460(0.0764)	0.0337(0.0093)	0.0592(0.0163)	9.2
OmniLMM-12B(24/4)	0.0412(0.0627)	0.0075(0.0232)	0.1339(0.1324)	0.1789(0.0330)	0.0173(0.0027)	0.0316(0.0050)	6.8
	0.0397(0.0614)	0.0067(0.0238)	0.1293(0.1336)	0.2095(0.0393)	0.0180(0.0040)	0.0331(0.0071)	7.3
VisualGLM-6B 23/5	0.0361(0.0519)	0.0182(0.0517)	0.1338(0.1214)	0.0662(0.0710)	0.0002(0.0002)	0.0004(0.0004)	4.3
	0.0306(0.0492)	0.0208(0.0588)	0.1173(0.1157)	0.3404(0.1494)	0.0014(0.0006)	0.0029(0.0012)	8.6
DeepSeek-VL2 (24/12)	0.2697(0.0675)	0.0939(0.0976)	0.4875(0.0536)	0.2170(0.0137)	0.1532(0.0076)	0.1795(0.0081)	23.4
	0.2817(0.0637)	0.1054(0.0974)	0.4936(0.0476)	0.2198(0.0269)	0.1571(0.0135)	0.1831(0.0176)	24.0
Medical-Specific VLMs							
MedDr(24/4)	0.2667(0.1012)	0.1564(0.0434)	0.4571(0.1168)	0.2270(0.0245)	0.1820(0.0201)	0.2020(0.0215)	24.9
	0.2801(0.0874)	0.1536(0.0389)	0.4742(0.0951)	0.2397(0.0275)	0.2113(0.0084)	0.2243(0.0138)	26.4
HuatuoGPT-Vision (24/6)	0.1384(0.0865)	0.0000(0.0000)	0.3399(0.1112)	0.1692(0.0232)	0.0814(0.0186)	0.1097(0.0207)	14.0
	0.2573(0.0546)	0.0743(0.0278)	0.4834(0.0577)	0.2183(0.0200)	0.1620(0.0148)	0.1859(0.0164)	23.0
MedVLM-R1 (25/2)	0.1602(0.1112)	0.0006(0.0097)	0.3472(0.1742)	0.2246(0.0285)	0.1019(0.0117)	0.1399(0.0150)	16.2
	0.1708(0.1294)	0.0003(0.0070)	0.3358(0.1840)	0.2321(0.0324)	0.1204(0.0077)	0.1583(0.0110)	17.0
MedGemma-4B (25/7)	0.3015(0.0517)	0.1215(0.0384)	0.5077(0.0385)	0.2276(0.0185)	0.2260(0.0113)	0.2266(0.0129)	26.8
	0.2874(0.0773)	0.1207(0.0339)	0.4875(0.0786)	0.2362(0.0162)	0.2245(0.0091)	0.2301(0.0103)	26.4
MedGemma-27B (25/7)	0.2185(0.0552)	0.0297(0.0157)	0.4390(0.0696)	0.2300(0.0375)	0.0391(0.0079)	0.0667(0.0130)	17.1
	0.2251(0.0574)	0.0309(0.0153)	0.4521(0.0781)	0.2853(0.0435)	0.0846(0.0141)	0.1304(0.0201)	20.1
Lingshu-7B (25/6)	0.2848(0.0855)	0.1079(0.0727)	0.4793(0.0933)	0.2281(0.0162)	0.1970(0.0100)	0.2112(0.0097)	25.1
	0.2775(0.0945)	0.1030(0.0748)	0.4700(0.1119)	0.2273(0.0220)	0.1942(0.0106)	0.2093(0.0138)	24.7
Lingshu-32B (25/6)	0.3050(0.0650)	0.1554(0.0520)	0.4999(0.0604)	0.2250(0.0178)	0.2035(0.0079)	0.2135(0.0109)	26.7
	0.2987(0.0713)	0.1531(0.0597)	0.4939(0.0733)	0.2328(0.0151)	0.2071(0.0071)	0.2191(0.0091)	26.8
Closed-Source VLMs							
Gemini 2.5 Pro (25/6)	0.1535(0.0411)	0.0186(0.0199)	0.3987(0.0438)	0.1705(0.0299)	0.0215(0.0056)	0.0381(0.0092)	13.4
	0.1536(0.0420)	0.0199(0.0201)	0.4025(0.0477)	0.2394(0.0571)	0.0311(0.0066)	0.0550(0.0115)	15.0
GPT-4o (24/5)	0.2023(0.0422)	0.0287(0.0160)	0.4023(0.0421)	0.3375(0.0891)	0.0527(0.0110)	0.0910(0.0185)	18.6
	0.2134(0.0425)	0.0318(0.0132)	0.4168(0.0412)	0.2540(0.0450)	0.0728(0.0085)	0.1130(0.0135)	18.5
Moonshot-v1 (25/1)	0.3064(0.0496)	0.1261(0.0301)	0.5157(0.0339)	0.2603(0.0220)	0.1457(0.0096)	0.1866(0.0117)	25.7
	0.2923(0.0464)	0.1055(0.0302)	0.5142(0.0316)	0.2327(0.0232)	0.1803(0.0132)	0.2030(0.0160)	25.5
Qwen-VL-Max (25/1)	0.2315(0.0375)	0.0269(0.0035)	0.4462(0.0377)	0.2764(0.0313)	0.1897(0.0139)	0.2248(0.0183)	23.3
	0.2479(0.0406)	0.0265(0.0047)	0.4649(0.0409)	0.2844(0.0220)	0.1802(0.0092)	0.2204(0.0113)	23.7

Table 1: Evaluation of general-purpose and medical-specific VLMs on PET2Rep benchmark. Evaluation results presented in gray and white represent the results of separate PET and CT images and fused PET/CT images, respectively.

a substantial challenge for existing VLMs. Many models are unable to consistently adhere to the prescribed report structure, occasionally generating disorganized, unusable, or even empty outputs, which yield near-zero scores across evaluation metrics and are thus omitted from the results table. Even when models attempt to follow the template, they often include irrelevant information or omit mandatory sections, underscoring their difficulty in capturing the core task requirements. This pattern suggests that many VLMs are overfitted to narrow training distributions, typically focused on specific tasks such as single-organ interpretation or generic image captioning, rather than holistic clinical reporting. As a result, they struggle to generalize to clinical applications, where accuracy, completeness, and structural consistency are essential.

State-of-the-Art Models Merely Match the Baseline.

Although the most advanced models, such as the Lingshu and MedGemma series, outperform other VLMs, their performance remains only marginally comparable to the baseline. This underwhelming result indicates that even state-of-the-art VLMs are far from ready for practical application in clinical workflow. While these models can generate coherent text with high NLG metrics, they frequently omit critical clinical details, such as subtle tracer uptake abnormalities, leading to low CE metrics. Manual review by radiologists further confirms that the outputs of these models are largely unusable. The accurate interpretation of tracer uptake patterns, combined with the extensive medical knowledge required for comprehensive whole-body assessment, remains a major challenge, highlighting the gap between general language proficiency and specialized clinical expertise. Further manual evaluation by two radiologists in Figure. 3 demonstrate that the outputs of state-of-the-art models are also mostly unusable. The nuanced interpretation of tracer uptake patterns and the broad medical knowledge required for whole-body assessment remain significant challenges, highlighting a critical gap between general language proficiency and specialized clinical expertise.

Larger Model Does Not Necessarily Translate to Better Performance. Our evaluation reveals an intriguing phenomenon that within the same model series, larger-scale models do not consistently outperform their smaller counterparts. In some cases, larger models appear to overlook task requirements, generating irrelevant or fabricated details such as patient names and ages, which negatively impact their evaluation performance. This observation suggests that the inferior performance of larger models may not stem from model scaling itself, but rather from insufficient exposure to domain-specific data and task-oriented training. Therefore, for specialized and highly structured tasks like PET report generation, architectural innovation and targeted fine-tuning may play a more critical role.

Further details regarding experimental results analysis and case studies are presented in the Appendix.

Discussion and Conclusion

In this work, we present PET2Rep, the first comprehensive benchmark specifically designed for evaluating radiology report generation in PET imaging, addressing a criti-

cal gap between existing research and clinical application. The benchmark consists of 565 whole-body PET/CT image-report cases, representing a significant advancement in this domain. Another key innovation is the introduction of a series of clinical efficacy metrics to evaluate the quality of radiotracer uptake pattern description in key organs in generated reports PET reports, which is a decisive factor in clinical decision-making, as omissions in critical findings can alter therapeutic pathways.

Our experimental results clearly reveal the critical limitations of current VLMs. Despite their reported success on various multimodal medical benchmarks, all models fail to surpass even a simple template baseline in PET2Rep, with some models generating disorganized or structurally non-compliant reports. These findings underscore the need for fundamental advancements in clinically grounded evaluation frameworks and rigorous alignment with real-world reporting standards to achieve genuine clinical applicability. Many existing benchmark tasks assess the capabilities of VLMs through visual question answering, which primarily reflects superficial image understanding and falls short of the deep clinical reasoning required for diagnosis and treatment. Moreover, existing clinical report generation datasets are largely confined to localized anatomical structures, overlooking the integration of structural and functional information necessary for comprehensive whole-body evaluation. In this context, PET2Rep serves as an expert-informed and clinically aligned benchmark that helps bridge this gap, providing a foundation for exploring the potential of large models toward more generalizable medical intelligence and facilitating progress in domain-specific model development.

While PET2Rep represents a significant step forward, several limitations should be acknowledged. At present, our evaluations are limited to 2D slices, which do not fully capture the three-dimensional spatial relationships and volumetric information critical for comprehensive image interpretation (Zhang et al. 2022). Moreover, clinically important quantitative indicators, such as standardized uptake values (SUVs) and lesion volume measurements, are not yet incorporated into the current evaluation framework. To address these limitations, we plan to expand the benchmark to support full 3D PET/CT evaluations, enabling more complete spatial and volumetric analysis (Xue et al. 2025). Key quantitative measures, including SUVs and lesion volumes, will be reintegrated to enhance the benchmark’s clinical validity. In addition, while the current version supports only Chinese reports, future iterations will extend to multilingual evaluation, improving generalizability and facilitating broader clinical adoption of VLMs across diverse healthcare systems (Qiu et al. 2024). Another consideration lies in the contextual limitations of current VLMs. Generating long, structured whole-body reports may exceed the effective context length or reasoning capacity of some models. In future work, we plan to further investigate this issue by exploring hierarchical strategies, such as summarizing findings by anatomical regions followed by integration into a standardized report template. These planned enhancements will significantly improve the benchmark’s clinical relevance and utility for developing more robust report generation systems.

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References

- Ashraf, N.; Tahir, M. J.; Saeed, A.; Ghosheh, M. J.; Alsheikh, T.; Ahmed, A.; Lee, K. Y.; and Yousaf, Z. 2023. Incidence and factors associated with burnout in radiologists: A systematic review. *European Journal of Radiology Open*, 11: 100530.
- Bai, J.; Bai, S.; Yang, S.; Wang, S.; Tan, S.; Wang, P.; Lin, J.; Zhou, C.; and Zhou, J. 2023. Qwen-vl: A frontier large vision-language model with versatile abilities. *arXiv preprint arXiv:2308.12966*.
- Bai, S.; Chen, K.; Liu, X.; Wang, J.; Ge, W.; Song, S.; Dang, K.; Wang, P.; Wang, S.; Tang, J.; et al. 2025. Qwen2. 5-vl technical report. *arXiv preprint arXiv:2502.13923*.
- Banerjee, S.; and Lavie, A. 2005. METEOR: An automatic metric for MT evaluation with improved correlation with human judgments. In *Proceedings of the acl workshop on intrinsic and extrinsic evaluation measures for machine translation and/or summarization*, 65–72.
- Bassi, P. R.; Yavuz, M. C.; Wang, K.; Chen, X.; Li, W.; Decherchi, S.; Cavalli, A.; Yang, Y.; Yuille, A.; and Zhou, Z. 2025. RadGPT: Constructing 3D Image-Text Tumor Datasets. *arXiv preprint arXiv:2501.04678*.
- Chen, J.; Gui, C.; Ouyang, R.; Gao, A.; Chen, S.; Chen, G. H.; Wang, X.; Zhang, R.; Cai, Z.; Ji, K.; et al. 2024. Huatuoogpt-vision, towards injecting medical visual knowledge into multimodal llms at scale. *arXiv preprint arXiv:2406.19280*.
- Chen, J.; Guo, H.; Yi, K.; Li, B.; and Elhoseiny, M. 2022. Visualgpt: Data-efficient adaptation of pretrained language models for image captioning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 18030–18040.
- Chen, Z.; Bie, Y.; Jin, H.; and Chen, H. 2025. Large language model with region-guided referring and grounding for ct report generation. *IEEE Transactions on Medical Imaging*.
- Coleman, R. E.; Hillner, B. E.; Shields, A. F.; Duan, F.; Merlino, D. A.; Hanna, L. G.; Stine, S. H.; and Siegel, B. A. 2010. PET and PET/CT reports: observations from the National Oncologic PET Registry. *Journal of Nuclear Medicine*, 51(1): 158–163.
- Comanici, G.; Bieber, E.; Schaekermann, M.; Pasupat, I.; Sachdeva, N.; Dhillon, I.; Blistein, M.; Ram, O.; Zhang, D.; Rosen, E.; et al. 2025. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*.
- Du, Z.; Qian, Y.; Liu, X.; Ding, M.; Qiu, J.; Yang, Z.; and Tang, J. 2022. GLM: General Language Model Pretraining with Autoregressive Blank Infilling. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 320–335.
- Feng, Y.; Liu, Y.; Yang, S.; Cai, W.; Zhang, J.; Zhan, Q.; Huang, Z.; Yan, H.; Wan, Q.; Liu, C.; et al. 2025. Vision-language model for object detection and segmentation: A review and evaluation. *arXiv preprint arXiv:2504.09480*.
- Gatidis, S.; Früh, M.; Fabritius, M. P.; Gu, S.; Nikolaou, K.; Fougère, C. L.; Ye, J.; He, J.; Peng, Y.; Bi, L.; et al. 2024. Results from the autoPET challenge on fully automated lesion segmentation in oncologic PET/CT imaging. *Nature Machine Intelligence*, 1–10.
- Hamamci, I. E.; Er, S.; and Menze, B. 2024. Ct2rep: Automated radiology report generation for 3d medical imaging. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, 476–486. Springer.
- Hamamci, I. E.; Er, S.; Wang, C.; Almas, F.; Simsek, A. G.; Esirgun, S. N.; Doga, I.; Durugol, O. F.; Dai, W.; Xu, M.; et al. 2024. Developing generalist foundation models from a multimodal dataset for 3d computed tomography. *arXiv preprint arXiv:2403.17834*.
- He, S.; Nie, Y.; Chen, Z.; Cai, Z.; Wang, H.; Yang, S.; and Chen, H. 2024. Meddr: Diagnosis-guided bootstrapping for large-scale medical vision-language learning. *arXiv e-prints*, arXiv:2404.
- Hurst, A.; Lerer, A.; Goucher, A. P.; Perelman, A.; Ramesh, A.; Clark, A.; Ostrow, A.; Welihinda, A.; Hayes, A.; Radford, A.; et al. 2024. Gpt-4o system card. *arXiv preprint arXiv:2410.21276*.
- Jiang, Y.; Omiye, J. A.; Zakka, C.; Moor, M.; Gui, H.; Alipour, S.; Mousavi, S. S.; Chen, J. H.; Rajpurkar, P.; and Daneshjou, R. 2024. Evaluating general vision-language models for clinical medicine. *medRxiv*, 2024–04.
- Jing, B.; Xie, P.; and Xing, E. 2018. On the Automatic Generation of Medical Imaging Reports. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2577–2586.
- Li, C.; Wong, C.; Zhang, S.; Usuyama, N.; Liu, H.; Yang, J.; Naumann, T.; Poon, H.; and Gao, J. 2023. Llava-med: Training a large language-and-vision assistant for biomedicine in one day. *Advances in Neural Information Processing Systems*, 36: 28541–28564.
- Lin, C.-Y. 2004. Rouge: A package for automatic evaluation of summaries. In *Text summarization branches out*, 74–81.
- Lin, H.; Hong, D.; Ge, S.; Luo, C.; Jiang, K.; Jin, H.; and Wen, C. 2025. Rs-moe: A vision-language model with mixture of experts for remote sensing image captioning and visual question answering. *IEEE Transactions on Geoscience and Remote Sensing*.
- Lin, H.; Xu, C.; and Qin, J. 2025. Taming Vision-Language Models for Medical Image Analysis: A Comprehensive Review. *arXiv preprint arXiv:2506.18378*.
- Liu, B.; Zou, K.; Zhan, L.; Lu, Z.; Dong, X.; Chen, Y.; Xie, C.; Cao, J.; Wu, X.-M.; and Fu, H. 2024. GEMeX:

- A Large-Scale, Groundable, and Explainable Medical VQA Benchmark for Chest X-ray Diagnosis. *arXiv preprint arXiv:2411.16778*.
- Liu, H.; Li, C.; Wu, Q.; and Lee, Y. J. 2023a. Visual instruction tuning. *Advances in neural information processing systems*, 36: 34892–34916.
- Liu, J.; Wang, Z.; Ye, Q.; Chong, D.; Zhou, P.; and Hua, Y. 2023b. Qilin-med-vl: Towards chinese large vision-language model for general healthcare. *arXiv preprint arXiv:2310.17956*.
- Lucignani, G.; Paganelli, G.; and Bombardieri, E. 2004. The use of standardized uptake values for assessing FDG uptake with PET in oncology: a clinical perspective. *Nuclear medicine communications*, 25(7): 651–656.
- Matsubara, K.; Ibaraki, M.; Nemoto, M.; Watabe, H.; and Kimura, Y. 2022. A review on AI in PET imaging. *Annals of Nuclear Medicine*, 36(2): 133–143.
- MoonshotAI. 2025. Moonshot model site. <https://www.moonshot.cn/>.
- Moor, M.; Huang, Q.; Wu, S.; Yasunaga, M.; Dalmia, Y.; Leskovec, J.; Zakka, C.; Reis, E. P.; and Rajpurkar, P. 2023. Med-flamingo: a multimodal medical few-shot learner. In *Machine Learning for Health (ML4H)*, 353–367. PMLR.
- Pan, J.; Liu, C.; Wu, J.; Liu, F.; Zhu, J.; Li, H. B.; Chen, C.; Ouyang, C.; and Rueckert, D. 2025. Medvlm-r1: Incentivizing medical reasoning capability of vision-language models (vlms) via reinforcement learning. *arXiv preprint arXiv:2502.19634*.
- Pang, T.; Li, P.; and Zhao, L. 2023. A survey on automatic generation of medical imaging reports based on deep learning. *BioMedical Engineering OnLine*, 22(1): 48.
- Papineni, K.; Roukos, S.; Ward, T.; and Zhu, W.-J. 2002. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, 311–318.
- Peng, C.; Zhang, K.; Lyu, M.; Liu, H.; Sun, L.; and Wu, Y. 2025. Scaling Up Biomedical Vision-Language Models: Fine-Tuning, Instruction Tuning, and Multi-Modal Learning. *arXiv preprint arXiv:2505.17436*.
- Peng, L.; Liao, Y.; Zhou, R.; Zhong, Y.; Jiang, H.; Wang, J.; Fu, Y.; Xue, L.; Zhang, X.; Sun, M.; et al. 2023. [18F] FDG PET/MRI combined with chest HRCT in early cancer detection: a retrospective study of 3020 asymptomatic subjects. *European Journal of Nuclear Medicine and Molecular Imaging*, 50(12): 3723–3734.
- Qiu, P.; Wu, C.; Zhang, X.; Lin, W.; Wang, H.; Zhang, Y.; Wang, Y.; and Xie, W. 2024. Towards building multilingual language model for medicine. *Nature Communications*, 15(1): 8384.
- Ren, S.; Laub, P.; Lu, Y.; Naganawa, M.; and Carson, R. E. 2019. Atlas-based multiorgan segmentation for dynamic abdominal PET. *IEEE Transactions on Radiation and Plasma Medical Sciences*, 4(1): 50–62.
- Schwenck, J.; Sonanini, D.; Cotton, J. M.; Rammensee, H.-G.; la Fougère, C.; Zender, L.; and Pichler, B. J. 2023. Advances in PET imaging of cancer. *Nature Reviews Cancer*, 23(7): 474–490.
- Sellergren, A.; Kazemzadeh, S.; Jaroensri, T.; Kiraly, A.; Traverse, M.; Kohlberger, T.; Xu, S.; Jamil, F.; Hughes, C.; Lau, C.; et al. 2025. MedGemma Technical Report. *arXiv preprint arXiv:2507.05201*.
- Sepehri, M. S.; Fabian, Z.; Soltanolkotabi, M.; and Soltanolkotabi, M. 2024. MediConfusion: Can you trust your AI radiologist? Probing the reliability of multimodal medical foundation models. *arXiv preprint arXiv:2409.15477*.
- Song, X.; Zhang, X.; Ji, J.; Liu, Y.; and Wei, P. 2022. Cross-modal contrastive attention model for medical report generation. In *Proceedings of the 29th International Conference on Computational Linguistics*, 2388–2397.
- van der Geest, K. S.; Treglia, G.; Glaudemans, A. W.; Brouwer, E.; Sandovici, M.; Jamar, F.; Gheysens, O.; and Slart, R. H. 2021. Diagnostic value of [18F] FDG-PET/CT for treatment monitoring in large vessel vasculitis: a systematic review and meta-analysis. *European journal of nuclear medicine and molecular imaging*, 48(12): 3886–3902.
- Wang, Y.; Lin, Z.; Xu, Z.; Dong, H.; Luo, J.; Tian, J.; Shi, Z.; Huang, L.; Zhang, Y.; Fan, J.; et al. 2024. Trust it or not: Confidence-guided automatic radiology report generation. *Neurocomputing*, 578: 127374.
- Wu, C.; Zhang, X.; Zhang, Y.; Wang, Y.; and Xie, W. 2023. Towards generalist foundation model for radiology by leveraging web-scale 2D&3D medical data. *arXiv preprint arXiv:2308.02463*.
- Wu, Z.; Chen, X.; Pan, Z.; Liu, X.; Liu, W.; Dai, D.; Gao, H.; Ma, Y.; Wu, C.; Wang, B.; et al. 2024. Deepseek-vl2: Mixture-of-experts vision-language models for advanced multimodal understanding. *arXiv preprint arXiv:2412.10302*.
- Xu, W.; Chan, H. P.; Li, L.; Aljunied, M.; Yuan, R.; Wang, J.; Xiao, C.; Chen, G.; Liu, C.; Li, Z.; et al. 2025. Lingshu: A Generalist Foundation Model for Unified Multimodal Medical Understanding and Reasoning. *arXiv preprint arXiv:2506.07044*.
- Xue, L.; Feng, G.; Zhang, W.; Zhang, Y.; Li, L.; Wang, S.; Peng, L.; Peng, S.; and Gao, X. 2025. PETWB-REP: A Multi-Cancer Whole-Body FDG PET/CT and Radiology Report Dataset for Medical Imaging Research. *arXiv preprint arXiv:2511.03194*.
- Xue, L.; Fu, Y.; Gao, X.; Feng, G.; Qian, S.; Wei, L.; Li, L.; Zhuo, C.; Zhang, H.; and Tian, M. 2024. [18F] FDG PET integrated with structural MRI for accurate brain age prediction. *European Journal of Nuclear Medicine and Molecular Imaging*, 51(12): 3617–3629.
- Ye, J.; Wang, G.; Li, Y.; Deng, Z.; Li, W.; Li, T.; Duan, H.; Huang, Z.; Su, Y.; Wang, B.; et al. 2024. Gmai-mmbench: A comprehensive multimodal evaluation benchmark towards general medical ai. *Advances in Neural Information Processing Systems*, 37: 94327–94427.
- Young, A.; Chen, B.; Li, C.; Huang, C.; Zhang, G.; Zhang, G.; Wang, G.; Li, H.; Zhu, J.; Chen, J.; et al. 2024. Yi: Open foundation models by 01. ai. *arXiv preprint arXiv:2403.04652*.

Yu, T.; Zhang, H.; Yao, Y.; Dang, Y.; Chen, D.; Lu, X.; Cui, G.; He, T.; Liu, Z.; Chua, T.-S.; et al. 2024. Rlaif-v: Aligning mllms through open-source ai feedback for super gpt-4v trustworthiness. *arXiv preprint arXiv:2405.17220*.

Zhang, J.; Huang, J.; Jin, S.; and Lu, S. 2024. Vision-language models for vision tasks: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.

Zhang, Y.; Liao, Q.; Ding, L.; and Zhang, J. 2022. Bridging 2D and 3D Segmentation Networks for Computation-Efficient Volumetric Medical Image Segmentation: An Empirical Study of 2.5 D Solutions. *Computerized Medical Imaging and Graphics*, 102088.

Zhang, Y.; Shen, Z.; and Jiao, R. 2024. Segment anything model for medical image segmentation: Current applications and future directions. *Computers in Biology and Medicine*, 108238.

Zhang, Y.; Xue, L.; Zhang, W.; Li, L.; Liu, Y.; Jiang, C.; Cheng, Y.; and Qi, Y. 2025. Seganypet: Universal promptable segmentation from positron emission tomography images. *arXiv preprint arXiv:2502.14351*.

Zhou, T.; Xu, Y.; Zhu, Y.; Xiao, C.; Bian, H.; Wei, L.; and Zhang, X. 2025. DrVD-Bench: Do Vision-Language Models Reason Like Human Doctors in Medical Image Diagnosis? *arXiv preprint arXiv:2505.24173*.

Zhu, J.; Wang, W.; Chen, Z.; Liu, Z.; Ye, S.; Gu, L.; Tian, H.; Duan, Y.; Su, W.; Shao, J.; et al. 2025a. Internvl3: Exploring advanced training and test-time recipes for open-source multimodal models. *arXiv preprint arXiv:2504.10479*.

Zhu, Q.; Hou, B.; Mathai, T. S.; Mukherjee, P.; Jin, Q.; Chen, X.; Wang, Z.; Cheng, R.; Summers, R. M.; and Lu, Z. 2025b. How well do multimodal LLMs interpret CT scans? An auto-evaluation framework for analyses. *Journal of Biomedical Informatics*, 168: 104864.

Appendix

A. Dataset Details

Data Collection. All patients fasted for at least six hours and had blood glucose levels below 11.1 mmol/L before the scan. Patients were intravenously administered [18F]FDG at a dose of 3.70–5.55 MBq/Kg and then rested for a 60-minute uptake period. Images were acquired using a Biograph 64 PET/CT scanner. The protocol included an initial CT scan (120 kV, 170 mA, 3.0 mm slice thickness) for attenuation correction, followed by a 3D PET scan over 5–6 bed positions with an acquisition time of 2.5 minutes per bed. Delayed imaging was performed in select cases. Final PET images were reconstructed using an iterative algorithm with CT-based attenuation correction.

Quality Control. All images were visually inspected by two experienced nuclear medicine physicians. Additionally, cases with severe artifacts such as motion, truncation, and metal artifacts are excluded.

Data Visualization. Visualization example cases of PET2Rep benchmark are shown in Figure. 4.

B. Evaluation Details

This appendix provides a detailed description of the implementation details of the evaluation metrics used for generated PET reports in our work.

Natural Language Generation Metrics. To quantitatively assess the quality of generated Chinese reports, we developed a dedicated evaluation pipeline tailored to the linguistic idiosyncrasies of the language. Initially, a rigorous preprocessing step is applied to both the generated and reference texts. This involves filtering the content to retain only Chinese characters, alphanumeric characters, and essential Chinese punctuation. Irrelevant special symbols are discarded to minimize noise while preserving semantic integrity. Given the absence of explicit word delimiters in written Chinese, we employ the Jieba segmentation library, a standard tool for Chinese word tokenization, to partition the character sequences into meaningful tokens. Any resulting empty or whitespace-only tokens are subsequently removed to ensure the fidelity of downstream calculations. For metric computation, we adapted standard n-gram-based metrics. The BLEU-1 through BLEU-4 scores are calculated based on the overlap of n-grams in the segmented token sequences. We utilize uniform weights for individual BLEU scores and incorporate a smoothing function to mitigate the impact of data sparsity, a common issue in shorter texts with limited token overlap. To evaluate structural correspondence, we compute the ROUGE-L score by applying the longest common subsequence to the tokenized outputs. This approach effectively captures the sequential and contextual alignment between the generated and reference reports, which is crucial for evaluating narrative coherence in Chinese.

PET Clinical Efficacy Metrics. Given that the whole-body PET imaging data used in our study, we extract the assessment of uptake levels corresponding to each key organ from the report text and compare these assessments with the

No.	Organ (Anatomic Structures)
1	Cranium and Brain
2	Eyeballs
3	Nasal Cavity and Sinuses
4	Pharynx and Parapharyngeal Space
5	Palatine Tonsils and Larynx
6	Salivary Glands and Thyroid
7	Cervical Lymph Nodes
8	Lungs and Thoracic Cavity
9	Mediastinum and Heart
10	Esophagus
11	Liver
12	Gallbladder
13	Pancreas
14	Spleen
15	Kidneys and Adrenal Glands
16	Gastrointestinal Tract
17	Prostate/Uterus and Bladder
18	Abdominal and Pelvic Cavities
19	Spine and Bones

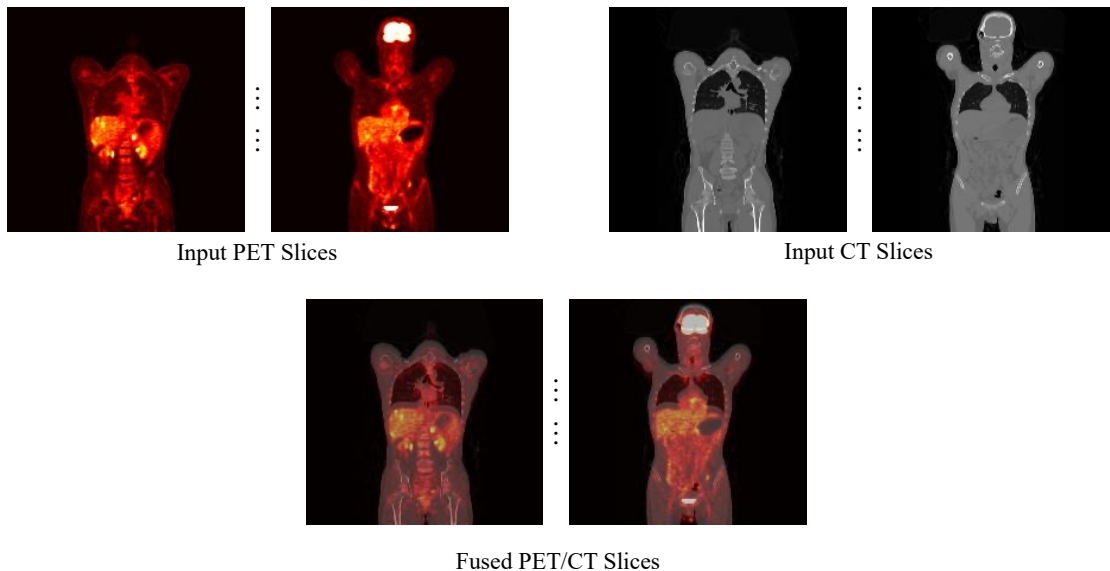
Table 2: List of key organs (anatomic structures) evaluated in PET Report clinical efficacy metrics.

corresponding entries in the ground truth reports. Our evaluation focuses on the assessment of radiotracer uptake for 19 predefined key organs and structures shown in Table. 2. For each key organ, we extract its uptake status from the report text and classify it into one of four mutually exclusive states of radiotracer uptake: *Increased Uptake*, *Decreased Uptake*, *Absent Uptake*, and *Normal*.

To ensure a consistent and fair comparison between generated and ground truth reports, we apply a set of normalization rules during the state extraction process for each of the 19 key organs. **(1) Default-to-Normal Assumption.** If a key organ is not explicitly mentioned in a report, its uptake status is automatically classified as normal. This rule reflects standard clinical reporting practice where only abnormal or clinically relevant normal findings are typically documented. **(2) Hierarchical Normality.** If a main organ category (e.g. "Lungs and Thoracic Cavity") is explicitly described as normal, all of its constituent sub-regions are inferred to be normal as well. **(3) Implicit Normality of Sub-regions.** If a report details a specific finding for a sub-region of a key organ (e.g. "increased uptake in subset X of the liver") but provides no information on other sub-regions, all other unmentioned sub-regions of that organ are classified as normal.

The evaluation protocol involves a multi-class classification assessment. For each of the 19 key organs, we compare the state assigned from the generated report to the state assigned from the ground truth report. The final CE metrics are calculated as the macro-average of the precision, recall, and F1-score across the three positive classes. For each positive class, we compute the True Positives (TPc), False Positives (FPc), and False Negatives (FNc) by aggregating counts across all 19 organs over the entire test set. The precision, recall, and F1-score for each class are then calculated

Example Case with Expert-Annotated Report



Original Report (Chinese): 禁食状态下静脉注射18F-FDG并静息后行全身PET/CT断层成像。全身断层影像显示：脑形态、结构正常，脑实质内未见异常密度影，FDG摄取未见异常。各脑室、脑沟、脑裂、脑池未见增宽，局部密度及FDG摄取未见异常，中线结构无偏移。双侧眼球形态和轮廓无异常，球后结构清晰，FDG摄取未见异常。副鼻窦黏膜未见明显增厚，鼻窦腔未见异常。鼻咽两侧软组织未见明显增厚，两侧咽隐窝对称，FDG摄取增高。舌根部及两侧扁桃体饱满，FDG摄取增高。喉咽形态、结构未见异常，咽旁间隙清晰。双侧腮腺、颌下腺大小、形态及密度未见异常，FDG呈生理性摄取。甲状腺形态、大小正常，未见明显异常密度影，FDG摄取未见异常。双侧颈深间隙、颌下及颌下未见明显肿大淋巴结，FDG摄取未见异常。两肺未见明显异常软组织结节及肿块影，FDG摄取未见异常。两侧胸腺未见明显增厚，两侧胸腔未见明显积液。心脏大小未见明显异常，心脏内密度低于心肌。输液港置入中。胃癌治疗后：胃充盈不良，胃体胃壁广泛性不均匀增厚，FDG摄取增高。贲门处胃壁较厚，FDG摄取增高。上腹部结构紊乱、显示不清，肝胃间隙、肝门区及腹膜后区见小及稍大淋巴结影，大者位于腹膜后。FDG摄取未见异常。左侧锁骨上窝及左上胸廓入口各见一低密度结节灶，前者较大，FDG摄取增高。余纵隔及两侧肺门未见明显肿大淋巴结影，FDG摄取未见明显增高。大网膜、肠系膜、两侧结肠旁沟及盆底腹膜稍增厚，以大网膜为著。内见一软组织结节灶，FDG摄取未见异常。盆腔少量积液。食管未见扩张，管壁未见明显增厚或肿块，FDG摄取未见增高。肠道充盈欠满意，局部未见肿块，部分肠管FDG摄取增高。肝脏形态、大小未见明显异常，肝缘光整，肝裂未见增宽。肝实质内见散在多发低密度结节及肿块影，边界尚清，部分形态不规则，大者，部分病灶内见斑片状稍高密度影，多数病灶FDG呈本底摄取，少数病灶FDG摄取增高。肝内外胆管未见扩张。胆囊形态大小未见异常，胆囊壁未见增厚，未见阳性结石和明显肿块，胆囊窝FDG摄取未见异常。胰腺周围间隙清楚，实质未见明显异常密度影，胰管未见增宽，FDG未见异常摄取。脾脏形态、大小基本正常，密度及FDG摄取未见异常。双侧肾上腺稍增粗，FDG摄取增高，以左侧为著。双肾形态、大小正常，肾实质未见明显异常密度影，FDG摄取未见明显异常。两侧肾盂肾盏及输尿管未见增宽，局部未见阳性结石影。前列腺形态及大小未见异常，FDG摄取未见局灶性异常增高。膀胱充盈较差，腔内未见阳性结石。椎体未见明显骨质异常，FDG未见异常摄取。
Translated English: Under fasting conditions, 18F-FDG was intravenously injected and then whole-body PET/CT tomography was performed after rest. The whole-body tomography showed that the brain morphology and structure were normal, no abnormal density shadows were observed in the brain parenchyma, and no abnormalities were found in FDG uptake. No widening was observed in the ventricles, sulci, fissures and cisterns. No abnormalities were found in local density and FDG uptake, and there was no deviation in the midline structure. The morphology and contour of both eyeballs were normal, the retrobulbar structure was clear, and no abnormalities were found in FDG uptake. The mucosa of the paranasal sinuses was not significantly thickened, and the sinus walls were intact. The soft tissues on both sides of the nasopharynx were not significantly thickened, and the pharyngeal crypts on both sides were symmetrical, with increased FDG uptake. The tonsils at the base of the tongue and on both sides of the palate are plump, and the uptake of FDG is increased. There were no abnormalities in the morphology and structure of the larynx and pharynx, and the parapharyngeal space was clear. The size, shape and density of bilateral parotid glands and submandibular glands were not abnormal, and FDG was uptake physiological. The morphology and size of the thyroid gland were normal, and no obvious abnormal density shadows were observed. No abnormalities were found in FDG uptake. No obvious enlarged lymph nodes were observed in the bilateral deep cervical Spaces, submandibular and submental areas, and no abnormalities were found in FDG uptake. No obvious abnormal soft tissue nodules or mass shadows were observed in both lungs, and no abnormalities were found in FDG uptake. No significant thickening was observed in the pleura on both sides, and no obvious effusion was found in the thoracic cavity on both sides. No obvious abnormalities were observed in the size of the heart, and the density in the heart cavity was lower than that of the myocardium. The infusion port is placed in. After gastric cancer treatment: Poor gastric filling, extensive and uneven thickening of the gastric body and wall. Increased FDG uptake. The gastric wall at the cardia is relatively thick, increasing FDG uptake. The upper abdominal structure is disordered and unclear. Small and slightly large lymph node shadows can be seen in the hepatogastric space, hilar area and retroperitoneal area, with the larger ones located behind the peritoneum. No abnormalities were observed in FDG uptake. A low-density nodule lesion was observed in the left supraclavicular fossa and at the entrance of the left upper thoracic cage respectively, with the former being larger. Increased FDG uptake. No obvious enlarged lymph node shadows were observed in the remaining mediastinum and both hila, and no significant increase in FDG uptake was observed. The omentum, mesentery, bilateral paracolic sulcus and pelvic floor peritoneum were slightly thickened, with the omentum being the most prominent. A soft tissue nodule was seen inside, and no abnormal FDG uptake was observed. A small amount of fluid accumulation in the pelvic cavity. No esophageal dilation was observed, no obvious thickening or mass was found on the tube wall, and no increase in FDG uptake was observed. The intestinal filling was unsatisfactory, no local masses were observed, and the FDG uptake in some intestinal tubes increased. No obvious abnormalities were observed in the shape and size of the liver. The liver margins were smooth and the liver fissures were not widened. Scattered and multiple low-density nodules and mass shadows are seen in the liver parenchyma, with relatively clear boundaries. Some have irregular shapes. In larger cases, patchy slightly high-density shadows are observed in some lesions. FDG uptake is shown in the background in most lesions, while in a few lesions, FDG uptake is increased. No dilation was observed in the intrahepatic and extrahepatic bile ducts. There were no abnormalities in the shape and size of the gallbladder, no thickening of the gallbladder wall, no positive stones or obvious masses, and no abnormalities in FDG uptake in the gallbladder fossa. The peripancreatic space is clear, no obvious abnormal density shadow is observed in the parenchyma, the pancreatic duct is not widened, and no abnormal uptake of FDG is observed. The morphology and size of the spleen were basically normal, and no abnormalities were found in density and FDG uptake. Both adrenal glands are slightly thickened, with increased FDG uptake, especially on the left side. The morphology and size of both kidneys were normal. No obvious abnormal density shadows were observed in the renal parenchyma, and no obvious abnormalities were found in FDG uptake. No widening was observed in the renal pelvis, calyces and ureters on both sides, and no positive stone shadows were found locally. There were no abnormalities in the morphology and size of the prostate, and no focal abnormal increase in FDG uptake was observed. The bladder was poorly filled and no positive stones were found in the lumen. No obvious bone structure abnormalities were observed in the vertebral bodies, and no abnormal uptake of FDG was found.

Expert-Annotated GT Report

Figure 4: An example case with expert-annotated radiology report.

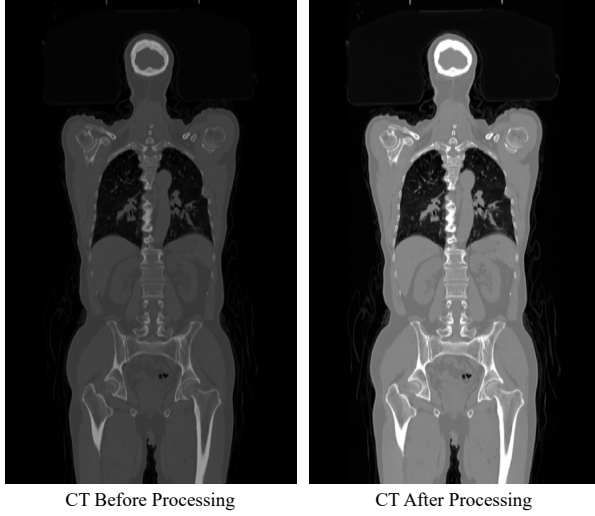


Figure 5: An example of CT image before and after preprocessing.

as follows:

$$\text{Precision}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c} \quad (1)$$

$$\text{Recall}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c} \quad (2)$$

$$\text{F1-score}_c = 2 \times \frac{\text{Precision}_c \times \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \quad (3)$$

Finally, the overall CE metrics are derived by macro-averaging the scores from the three positive classes. These macro-averaged scores provide a balanced measure of the model’s ability to correctly identify the presence and type of clinically significant abnormalities across all key organs.

C. Preprocessing Details

CT Resampling and Normalization. In PET/CT imaging, a key pre-processing step involves resampling the CT images to match the lower spatial resolution of the PET images. This coregistration process ensures that both imaging modalities share the same matrix size and that the voxels in each dataset correspond to the same anatomical location. By aligning the spatial resolution and dimensions of the CT and PET images, we eliminate potential confounding factors that could arise from their different original acquisition parameters. This alignment is critical for accurate multimodal analysis. After resampling, the intensity of the CT images is standardized using z-score. An example of PET image before and after preprocessing is shown in Figure. 5.

PET SUV Normalization. Another vital preprocessing step is the normalization of the PET data. The raw PET images, which measure radioactivity in counts, are converted to Standardized Uptake Values (SUV), which is a widely accepted quantitative metric in PET imaging. This conversion is significant because it allows for meaningful, standardized comparisons of tracer uptake across different patients and at

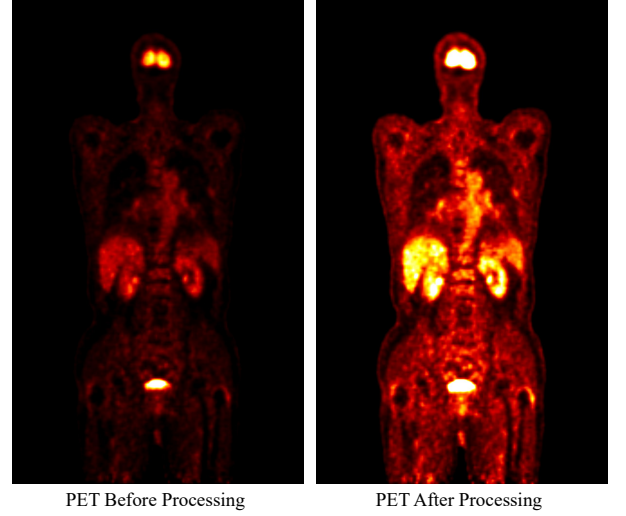


Figure 6: An example of PET image before and after preprocessing.

different times. The SUV is then calculated by dividing the tissue radioactivity concentration by the normalized injection dose as follows.

$$\text{SUV} = \frac{C(t)}{\text{Dose}_{\text{norm}}} = \frac{\text{RC (kBq/mL)} \cdot 2^{t/T}}{\text{ID (MBq)}/\text{BW (kg)}} \quad (4)$$

where RC represents the tissue radioactivity concentration, ID represents the injected dose, BW represents body weight.

An example of PET image before and after preprocessing is shown in Figure. 5.

Creation of Fused PET/CT Images. To emulate the clinical diagnostic workflow, we implement a process to fuse PET and CT scans. This technique combines the functional information from PET with the anatomical detail from CT, mirroring how radiologists interpret these images for diagnosis. The fused image is created by superimposing the pseudo-colored PET image, which illustrates metabolic activity, onto the corresponding grayscale CT image that provides the anatomical framework. This process yields a single composite view for visualizing functional information within its precise anatomical context. This integration is critical for accurately localizing areas of abnormal radiotracer uptake and enables a more holistic and integrated assessment of the patient’s condition.

D. Additional Experiments and Discussion

Manual Evaluation by Human Experts. To complement the automated quantitative metrics, we conduct a qualitative clinical evaluation to assess the practical utility and diagnostic reliability of the generated reports. For this assessment, we select the outputs from the highest-performing VLMs as determined by NLG and CE metrics. These selected model-generated reports alongside the corresponding ground truth reports are anonymized and presented to two radiologists for a blind review. The physicians are tasked with scoring each

Evaluation Dimension	Score	Meaning
Medical Accuracy	1	Completely inaccurate, with obvious errors (e.g., incorrect description of lesion location/nature).
	2	Multiple inaccuracies exist; severe issues in the overall description.
	3	Some details are inaccurate, but the overall judgment is reasonable.
	4	Only minor inaccuracies, acceptable.
	5	Completely accurate, highly consistent with image features.
Key Findings Completeness	1	Serious omission of important findings (e.g., missed reporting of the main lesion).
	2	Multiple important findings are omitted.
	3	The report is basically complete, but with minor omissions.
	4	Findings are comprehensive, with only minor omissions.
	5	All important findings are accurately presented, with no omissions.
Expression Clarity	1	Expression is chaotic, terminology is inappropriate, difficult to understand.
	2	Expression is unclear, with significant language issues.
	3	Basically coherent, with some grammatical errors or unprofessional terms.
	4	Expression is clear and professional, with only a few suboptimal word choices.
	5	Expression is precise, logically coherent, and conforms to standard radiological reporting style.
Clinical Usability	1	Completely unusable for clinical decision-making, needs to be rewritten.
	2	Report has many issues, requires extensive revisions before use.
	3	Can be used for reference, but parts need to be revised.
	4	Basically usable, only requires minor polishing.
	5	Can be directly used as a formal clinical report.
Overall Rating	1	Overall quality is poor, unacceptable.
	2	Has obvious problems, not recommended for use.
	3	Acceptable, but needs improvement.
	4	Good performance, close to clinically usable standard.
	5	High-quality generation, nearly flawless.

Table 3: Detailed multi-dimensional evaluation criteria of manual experts for report quality.

generated report based on the following dimensions, including Medical Accuracy (MedAcc), Key Findings Completeness (FinCom), Expression Clarity (ExpCla), Clinical Usability (CliUsa) and Overall Rating (OveRat). The scores range from 1 to 5 following the criteria in Table. 3, with higher scores indicating better performance.

From manual evaluation results shown in Figure. 3, where a larger and more outward-reaching polygon on the chart signifies higher scores across the evaluation dimensions, we can observe that moonshot-v1 is the best-performing model among the three VLMs for comparison. Comparing the two charts on the left (Expert A) with the two on the right (Expert B), the evaluation trends are highly consistent. How-

ever, none of the models have achieved scores that meet the stringent requirements of clinical practice, falling short of the thresholds necessary for reliable clinical application.

E. Case Study

In this section, we present a case study analysis of several VLMs in the PET2Rep benchmark, focusing on identifying and dissecting the key failure modes exhibited by the models during their performance. We classify the types of failure cases along the input-to-output pipeline into the following distinct categories, each characterized by specific behavioral patterns and illustrated with corresponding examples in the figures.

Model (year/month)	NLG Metrics						CE Metrics		
	BL-1	BL-2	BL-3	BL-4	MTR	RG-L	Pre	Rec	F1
Template Baseline	0.6026(0.0463)	0.4668(0.0455)	0.3851(0.0465)	0.3150(0.0482)	0.1475(0.0141)	0.5110(0.0319)	0.2282(0.0179)	0.2220(0.0106)	0.2249(0.0123)
General-Purpose VLMs									
Qwen2.5-VL-7B (25/1)	0.5957(0.0469)	0.4592(0.0455)	0.3762(0.0460)	0.3050(0.0476)	0.1407(0.0198)	0.5075(0.0340)	0.2233(0.0236)	0.1974(0.0083)	0.2094(0.0132)
	0.5975(0.0447)	0.4606(0.0437)	0.3772(0.0447)	0.3057(0.0467)	0.1390(0.0186)	0.5088(0.0320)	0.2284(0.0227)	0.2023(0.0075)	0.2144(0.0121)
Qwen2.5-VL-32B (25/1)	0.4361(0.0557)	0.3176(0.0506)	0.2428(0.0467)	0.1777(0.0421)	0.0063(0.0110)	0.4165(0.0516)	0.3402(0.0781)	0.0418(0.0127)	0.0743(0.0214)
	0.4468(0.0526)	0.3267(0.0484)	0.2511(0.0445)	0.1851(0.0408)	0.0063(0.0111)	0.4295(0.0486)	0.2728(0.0447)	0.0308(0.0047)	0.0554(0.0082)
Qwen2.5-VL-72B (25/1)	0.4718(0.0896)	0.3591(0.0754)	0.2855(0.0657)	0.2223(0.0585)	0.0655(0.0172)	0.4234(0.0588)	0.2474(0.0513)	0.0295(0.0024)	0.0527(0.0043)
	0.4854(0.0867)	0.3683(0.0732)	0.2923(0.0645)	0.2273(0.0584)	0.0645(0.0171)	0.4306(0.0594)	0.2917(0.0328)	0.0393(0.0049)	0.0693(0.0084)
InternVL3-8B (25/4)	0.5027(0.0736)	0.3767(0.0683)	0.3048(0.0654)	0.2439(0.0627)	0.0606(0.0443)	0.4739(0.0630)	0.2425(0.0151)	0.2107(0.0114)	0.2254(0.0119)
	0.5112(0.0695)	0.3846(0.0607)	0.3122(0.0560)	0.2509(0.0529)	0.0641(0.0463)	0.4845(0.0566)	0.2333(0.0153)	0.2099(0.0074)	0.2208(0.0087)
InternVL3-14B (25/4)	0.5141(0.1088)	0.3889(0.0883)	0.3149(0.0770)	0.2513(0.0684)	0.0472(0.0528)	0.4835(0.0910)	0.2366(0.0206)	0.2057(0.0095)	0.2199(0.0129)
	0.5126(0.1080)	0.3873(0.0875)	0.3132(0.0760)	0.2495(0.0671)	0.0532(0.0506)	0.4813(0.0904)	0.2322(0.0196)	0.1982(0.0099)	0.2137(0.0131)
InternVL3-38B (25/4)	0.2870(0.1789)	0.2156(0.1379)	0.1741(0.1129)	0.1377(0.0924)	0.0775(0.0483)	0.4371(0.1199)	0.2711(0.0203)	0.2072(0.0141)	0.2344(0.0127)
	0.3001(0.1643)	0.2258(0.1271)	0.1828(0.1043)	0.1446(0.0855)	0.0825(0.0480)	0.4618(0.0913)	0.2674(0.0258)	0.2435(0.0298)	0.2546(0.0278)
InternVL3-78B (25/4)	0.5765(0.0652)	0.4565(0.0585)	0.3762(0.0546)	0.3090(0.0525)	0.1233(0.0359)	0.4997(0.0401)	0.2355(0.0255)	0.0520(0.0119)	0.0850(0.0157)
	0.5787(0.0645)	0.4571(0.0576)	0.3776(0.0538)	0.3090(0.0518)	0.1262(0.0318)	0.5008(0.0397)	0.2369(0.0492)	0.0748(0.0083)	0.1132(0.0122)
Yi-VL-6B (24/1)	0.0185(0.0716)	0.0124(0.0519)	0.0091(0.0406)	0.0065(0.0316)	0.0002(0.0056)	0.0479(0.0709)	0.1144(0.0430)	0.0061(0.0020)	0.0115(0.0038)
	0.0998(0.1673)	0.0688(0.1215)	0.0515(0.0953)	0.0374(0.0733)	0.0029(0.0165)	0.1156(0.1432)	0.1519(0.0261)	0.0260(0.0033)	0.0464(0.0055)
Yi-VL-34B (24/1)	0.5144(0.1847)	0.3949(0.1472)	0.3227(0.1253)	0.2610(0.1071)	0.0848(0.0664)	0.4439(0.1420)	0.2305(0.0159)	0.1869(0.0079)	0.2063(0.0098)
	0.5592(0.1245)	0.4297(0.1029)	0.3518(0.0907)	0.2854(0.0809)	0.0898(0.0645)	0.4779(0.0950)	0.2303(0.0211)	0.2038(0.0072)	0.2160(0.0116)
LLaVa-V1.5-7B (23/9)	0.2856(0.0864)	0.2061(0.0721)	0.1581(0.0611)	0.1198(0.0508)	0.0126(0.0515)	0.3043(0.0639)	0.2044(0.0287)	0.1022(0.0091)	0.1306(0.0121)
	0.0915(0.0307)	0.0607(0.0223)	0.0446(0.0175)	0.0328(0.0141)	0.0056(0.0369)	0.1717(0.0283)	0.2460(0.0764)	0.0337(0.0093)	0.0592(0.0163)
OmniLMM-12B(24/4)	0.1275(0.1559)	0.815(0.1097)	0.580(0.0830)	0.0412(0.0627)	0.0075(0.0232)	0.1339(0.1324)	0.1789(0.0330)	0.0173(0.0027)	0.0316(0.0050)
	0.1253(0.1546)	0.0788(0.1077)	0.0560(0.0811)	0.0397(0.0614)	0.0067(0.0238)	0.1293(0.1336)	0.2095(0.0393)	0.0180(0.0040)	0.0331(0.0071)
VisualGLM-6B 23/5	0.1260(0.1523)	0.0771(0.0995)	0.0530(0.0719)	0.0361(0.0519)	0.0182(0.0517)	0.1338(0.1214)	0.0662(0.0710)	0.0002(0.0032)	0.0004(0.0004)
	0.1055(0.1427)	0.0644(0.0932)	0.0446(0.0680)	0.0306(0.0492)	0.0208(0.0588)	0.1173(0.1157)	0.3404(0.1494)	0.0014(0.0006)	0.0029(0.0012)
DeepSeek-VL2 (24/12)	0.5531(0.0760)	0.4181(0.0722)	0.3379(0.0697)	0.2697(0.0675)	0.0939(0.0976)	0.4875(0.0536)	0.2170(0.0137)	0.1532(0.0076)	0.1795(0.0081)
	0.5684(0.0663)	0.4322(0.0648)	0.3510(0.0641)	0.2817(0.0637)	0.1054(0.0974)	0.4936(0.0476)	0.2198(0.0269)	0.1571(0.0135)	0.1831(0.0176)
Medical-Specific VLMs									
MedDr(24/4)	0.5246(0.1732)	0.4029(0.1385)	0.3288(0.1180)	0.2667(0.1012)	0.1564(0.0434)	0.4571(0.1168)	0.2270(0.0245)	0.1820(0.0201)	0.2020(0.0215)
	0.5495(0.1427)	0.4225(0.1155)	0.3451(0.0999)	0.2801(0.0874)	0.1536(0.0389)	0.4742(0.0951)	0.2397(0.0275)	0.2113(0.0084)	0.2243(0.0138)
HuatuoGPT-Vision (24/6)	0.3748(0.1444)	0.2526(0.1195)	0.1876(0.1022)	0.1384(0.0865)	0.0000(0.0000)	0.3399(0.1112)	0.1692(0.0232)	0.0814(0.0186)	0.1097(0.0207)
	0.5443(0.0695)	0.4097(0.0623)	0.3284(0.0579)	0.2573(0.0546)	0.0743(0.0278)	0.4834(0.0577)	0.2183(0.0200)	0.1620(0.0148)	0.1859(0.0164)
MedVLM-R1 (25/2)	0.3673(0.1888)	0.2642(0.1567)	0.2069(0.1330)	0.1602(0.1112)	0.0006(0.0097)	0.3472(0.1742)	0.2246(0.0285)	0.1019(0.0117)	0.1399(0.0150)
	0.3738(0.2159)	0.2723(0.1805)	0.2160(0.1541)	0.1708(0.1294)	0.0003(0.0070)	0.3358(0.1840)	0.2321(0.0324)	0.1204(0.0077)	0.1583(0.0110)
MedGemma-4B (25/7)	0.5811(0.0621)	0.4496(0.0553)	0.3700(0.0527)	0.3015(0.0517)	0.1215(0.0384)	0.5077(0.0385)	0.2276(0.0185)	0.2260(0.0113)	0.2266(0.0129)
	0.5615(0.1118)	0.4321(0.0952)	0.3542(0.0853)	0.2874(0.0773)	0.1207(0.0339)	0.4875(0.0786)	0.2362(0.0162)	0.2245(0.0091)	0.2301(0.0103)
MedGemma-27B (25/7)	0.4748(0.0766)	0.3535(0.0676)	0.2801(0.0612)	0.2185(0.0552)	0.0297(0.0157)	0.4390(0.0696)	0.2300(0.0375)	0.0391(0.0079)	0.0667(0.0130)
	0.4881(0.0861)	0.3631(0.0737)	0.2878(0.0652)	0.2251(0.0574)	0.0309(0.0153)	0.4521(0.0781)	0.2853(0.0435)	0.0846(0.0141)	0.1304(0.0201)
Lingshu-7B (25/6)	0.5566(0.1294)	0.4278(0.1083)	0.3505(0.0960)	0.2848(0.0855)	0.1079(0.0727)	0.4793(0.0933)	0.2281(0.0162)	0.1970(0.0100)	0.2112(0.0097)
	0.5432(0.1519)	0.4173(0.1242)	0.3418(0.1081)	0.2775(0.0945)	0.1030(0.0748)	0.4700(0.1119)	0.2273(0.0220)	0.1942(0.0106)	0.2093(0.0138)
Lingshu-32B (25/6)	0.5852(0.0879)	0.4530(0.0757)	0.3732(0.0695)	0.3050(0.0650)	0.1554(0.0520)	0.4999(0.0604)	0.2250(0.0178)	0.2035(0.0079)	0.2135(0.0109)
	0.5761(0.1059)	0.4450(0.0881)	0.3661(0.0787)	0.2987(0.0713)	0.1531(0.0597)	0.4939(0.0733)	0.2328(0.0151)	0.2071(0.0071)	0.2191(0.0091)
Closed-Source VLMs									
Gemini 2.5 Pro (25/6)	0.4185(0.0530)	0.2786(0.0479)	0.2064(0.0447)	0.1535(0.0411)	0.0186(0.0199)	0.3987(0.0438)	0.1705(0.0299)	0.0215(0.0056)	0.0381(0.0092)
	0.4157(0.0602)	0.2780(0.0508)	0.2062(0.0463)	0.1536(0.0420)	0.0199(0.0201)	0.4025(0.0477)	0.2394(0.0571)	0.0311(0.0066)	0.0550(0.0115)
GPT-4o (24/5)	0.4534(0.0695)	0.3401(0.0552)	0.2658(0.0475)	0.2023(0.0422)	0.0287(0.0160)	0.4023(0.0421)	0.3375(0.0891)	0.0527(0.0110)	0.0910(0.0185)
	0.4792(0.0635)	0.3570(0.0520)	0.2798(0.0462)	0.2134(0.0425)	0.0318(0.0132)	0.4168(0.0412)	0.2540(0.0450)	0.0728(0.0085)	0.1130(0.0135)
Moonshot-v1 (25/1)	0.5888(0.0604)	0.4567(0.0545)	0.3754(0.0511)	0.3064(0.0496)	0.1261(0.0301)	0.5157(0.0339)	0.2603(0.0220)	0.1457(0.0096)	0.1866(0.0117)
	0.5655(0.0585)	0.4373(0.0518)	0.3594(0.0480)	0.2923(0.0464)	0.1055(0.0302)	0.5142(0.0316)	0.2327(0.0232)	0.1803(0.0132)	0.2030(0.0160)
Qwen-VL-Max (25/1)	0.5061(0.0457)	0.3793(0.0403)	0.3003(0.0375)	0.2315(0.0375)	0.0269(0.0035)	0.4462(0.0377)	0.2764(0.0313)	0.1897(0.0139)	0.2248(0.0183)
	0.5309(0.0507)	0.3996(0.0441)	0.3180(0.0416)	0.2479(0.0406)	0.0265(0.0047)	0.4649(0.0409)	0.2844(0.0220)	0.1802(0.0092)	0.2204(0.0113)

Table 4: Detailed evaluation results of PET2Rep benchmark. Evaluation results presented in gray and white represent the results of separate PET and CT images and fused PET/CT images, respectively.

Normal Outputs Following the Template. This category refers to instances where VLMs generate responses that strictly adhere to the structure and requirements of the provided report template, while effectively incorporating relevant information from the input data. Such outputs demonstrate the models’ ability to process the input accurately and produce coherent, task-appropriate content, as shown in the examples in Figure. 7 and Figure. 8.

Irrelevant Information. VLMs in this category generate responses that include content unrelated to the input data or the requirements of the report template. These outputs may contain extraneous details, off-topic information, or data not pertinent to the specific case, as illustrated in Figure 9.

Unstructured Outputs. Unstructured outputs without following the report template are those that completely disregard the predefined structure and format of the required report. Unlike chaotic outputs which are often disorganized and nonsensical, these outputs may contain some relevant information related to the input data but fail to present it in a structured manner, as illustrated in Figure 10.

Refuse to Answer. This category encompasses instances where VLMs explicitly decline to generate a response to the given input, often citing reasons related to involving sensitive or ethical issues, as illustrated in Figure 11.

Chaotic / Empty Outputs. Chaotic outputs are those where VLMs produce words or sentences that are unrelated to the image information. These outputs are not only irrelevant but also lack any logical connection to the input PET data or the requirements of the report template. While some VLMs produce no meaningful content at all, resulting in completely empty outputs. Such failures indicate a total inability to process the input and generate a response, which are exemplified in Figure 11. Due to the complete unavailability of the results and the fact that the evaluation metrics are almost all zero, we have removed the results of these VLMs from the comparison tables in the main text.

Case Study ——— Output Following the Template Structure

Original Response (Chinese): 禁食状态下静脉注射18F-FDG，静息60分钟后行全身PET断层显像。全身断层影像显示：脑形态、结构正常，双侧大脑皮质放射性分布均匀、对称，FDG摄取未见明显异常。各脑室、脑沟、脑裂、脑池形态正常，两侧脑室对称，中线结构无偏移。双侧眼球形态和轮廓无异常，FDG无异常摄取。鼻窦未见异常，FDG摄取未见异常。鼻咽壁未见增厚，两侧腭扁桃体对称，FDG呈生理性摄取。咽喉形态、结构未见异常。双侧腮腺、颌下腺形态、密度正常，FDG生理性摄取。甲状腺形态、大小正常，密度均匀，FDG摄取未见异常。双侧颈深间隙、颌下见数枚小淋巴结显示，部分淋巴结FDG摄取增高。两肺透亮度正常，两肺内未见明显异常密度影，FDG摄取未见异常。心影未见异常，部分动脉管壁钙化（包括冠状动脉）。食管未见扩张，管壁未见明显增厚或肿块，FDG摄取未见增高。肝缘光整，肝裂未见明显增厚，CT平扫肝右叶见一较大低密度影，边界清晰，FDG摄取增高。肝左叶未见异常密度影，FDG摄取正常。门静脉主干及左右支未见明显异常，FDG摄取正常。门腔间隙、肝胃间隙、腹膜后未见异常淋巴结显示，FDG摄取未见异常。胆囊大小形态正常，胆囊壁未见增厚，局部FDG摄取未见异常。胰腺形态正常，未见明显异常密度影，FDG摄取未见明显异常。脾脏形态、大小密度未见异常，FDG摄取未见异常。双侧泌尿系统见对比剂残留影；双肾形态大小正常，未见异常密度影，FDG摄取未见异常。双侧肾盂肾盏及输尿管未见增宽。双侧肾上腺显影未见明显异常，FDG摄取未见异常。胃充盈尚可，胃壁未见明显增厚，部分胃壁FDG摄取正常。肠道充盈欠佳，肠壁未见明显增厚或肿块，FDG呈生理性摄取。前列腺大小形态正常，实质内未见明显异常密度影，FDG摄取未见异常增高。膀胱充盈欠佳，内见对比剂残留影。骨质密度正常，脊柱未见明显异常。FDG摄取未见异常。 Translated English: Under fasting conditions, 18F-FDG was intravenously injected and then whole-body PET/CT tomography was performed after rest. After resting for 60 minutes, whole-body PET tomographic imaging was performed. The whole-body tomographic imaging showed that the brain morphology and structure were normal, the radioactive distribution of the bilateral cerebral cortex was uniform and symmetrical, and no obvious abnormalities were observed in FDG uptake. The morphology of each ventricle, sulcus, fissure and cisterns is normal. The bilateral ventricles are symmetrical, and there is no deviation in the midline structure. There were no abnormalities in the morphology and contour of both eyes, and no abnormal uptake of FDG. The mucosa of the right maxillary sinus was slightly thickened with partial calcification, and no abnormalities were observed in FDG uptake. The nasopharyngeal wall was not thickened, the palatine tonsils on both sides were symmetrical, and FDG was physiologically taken up. No abnormalities were found in the morphology and structure of the larynx and pharynx. The morphology and density of bilateral parotid glands and submandibular glands are normal, and FDG physiological uptake occurs. The morphology and size of the thyroid gland were normal, with uniform density. No abnormalities were observed in FDG uptake. Several small lymph nodes were found in the deep cervical spaces and submandibular areas on both sides, and FDG uptake was increased in some lymph nodes. The transparency of both lungs was normal. No obvious abnormal density shadows were observed in the lung fields, and no abnormalities were found in FDG uptake. No abnormalities were found in the cardiac shadow, but some arterial walls were calcified (including the coronary arteries). No esophageal dilation was observed, no obvious thickening or mass was found on the tube wall, and no increase in FDG uptake was observed. The liver margin was smooth and neat, and no significant widening of the liver fissure was observed. On CT plain scan, a large low-density shadow was seen in the right lobe of the liver with a clear boundary, and FDG uptake was increased. No abnormal density shadow was observed in the left lobe of the liver, and FDG uptake was normal. No obvious abnormalities were found in the main portal vein and its left and right branches, and the FDG uptake was normal. No abnormal lymph nodes were found in the portal space, hepatogastric space or retroperitoneum, and no abnormalities were observed in FDG uptake. The size and shape of the gallbladder are normal. There is no thickening of the gallbladder wall, and no abnormality in local FDG uptake is observed. The pancreatic morphology was normal, with no obvious abnormal density shadows observed, and no obvious abnormalities in FDG uptake were found. No abnormalities were observed in the morphology, size and density of the spleen, and no abnormalities were found in FDG uptake. Contrast agent residual images were observed in both urinary systems. The shape and size of both kidneys were normal, no abnormal density shadows were observed, and no abnormalities were found in FDG uptake. No widening was observed in the bilateral renal pelvis, calyces and ureters. No obvious abnormalities were found in bilateral adrenal imaging, and no abnormalities were observed in FDG uptake. The stomach was moderately filled, with no significant thickening of the stomach wall. FDG uptake in some parts of the stomach wall was normal. The intestinal filling was poor, and no obvious thickening or mass was observed in the intestinal wall. FDG was taken up physiologically. The size and shape of the prostate were normal. No obvious abnormal density shadows were observed in the parenchyma, and there was no abnormal increase in FDG uptake. The bladder was not well filled, and a residual image of contrast agent was seen inside. The bone density is normal and no obvious abnormalities are found in the spine. No abnormalities were observed in FDG uptake.
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Generated Report (InternVL3-78B)

Original Response (Chinese): 禁食状态下静脉注射18F-FDG，静息60分钟后行全身PET断层显像。全身断层影像显示：脑形态、结构正常，双侧大脑皮质放射性分布均匀、对称，FDG摄取未见明显异常。各脑室、脑沟、脑裂、脑池形态增宽，两侧脑室对称，中线结构无偏移。双侧眼球形态和轮廓无异常，FDG无异常摄取。右侧上颌窦黏膜稍增厚伴部分钙化，FDG摄取未见异常。鼻咽壁未见增厚，两侧腭扁桃体对称，FDG呈生理性摄取。咽喉形态、结构未见异常。双侧腮腺、颌下腺形态、密度正常，FDG生理性摄取。甲状腺形态、大小正常，密度均匀，FDG摄取未见异常。双侧颈深间隙、颌下见数枚小淋巴结显示，部分淋巴结FDG摄取增高。两肺透亮度增加，两肺上叶胸膜下见囊状透亮影。右肺中叶体积缩小，右肺中叶肺门旁见片状密度增高影，FDG摄取增高。心影未见异常。部分动脉管壁钙化（包括冠状动脉）。食管未见扩张，管壁未见明显增厚或肿块，FDG摄取未见增高。肝缘欠光整，肝裂未见明显增宽，肝右叶下段见巨大稍低密度肿块影，边界欠清，FDG摄取增高。肝左叶见稍低密度结节，边界尚清晰，FDG呈生理性摄取。肝右叶膈顶区另见结节状致密影。门静脉主干及左右支明显增粗，密度增高，FDG摄取增高。门腔间隙、肝胃间隙、腹膜后见淋巴结显示，FDG摄取轻度增高。胆囊稍饱满，胆囊壁未见增厚，局部FDG摄取未见异常。胰腺形态、大小密度未见异常，FDG摄取未见明显异常。脾脏形态、大小密度未见异常，FDG摄取未见异常。双侧泌尿系统见对比剂残留影；左肾体积缩小，左肾见多枚囊状低密度影及稍高密度小结节，边界清，FDG摄取缺失；右肾上腺极见一脂肪密度结节，FDG摄取未见异常。双侧肾盂肾盏及输尿管未见增宽。双侧肾上腺显影未见明显异常，FDG摄取未见异常增高。胃充盈尚可，胃壁未见明显增厚，部分胃壁FDG摄取增高。肠道充盈欠佳，肠壁未见明显增厚或肿块，FDG呈生理性摄取。前列腺稍增大，实质内见数枚钙化灶，FDG摄取未见异常增高。膀胱充盈欠佳，内见对比剂残留影。骨质密度不均匀减低，脊柱部分椎体边缘骨质增生，FDG摄取未见异常。 Translated English: Under fasting conditions, 18F-FDG was intravenously injected and then whole-body PET/CT tomography was performed after rest. After resting for 60 minutes, whole-body PET tomographic imaging was performed. The whole-body tomographic imaging showed that the brain morphology and structure were normal, the radioactive distribution of the bilateral cerebral cortex was uniform and symmetrical, and no obvious abnormalities were observed in FDG uptake. The morphology of each ventricle, sulcus, fissures and cisterns is widened, the bilateral ventricles are symmetrical, and the midline structure has no deviation. There were no abnormalities in the morphology and contour of both eyes, and no abnormal uptake of FDG. The mucosa of the right maxillary sinus was slightly thickened with partial calcification, and no abnormalities were observed in FDG uptake. The nasopharyngeal wall was not thickened, the palatine tonsils on both sides were symmetrical, and FDG was physiologically taken up. No abnormalities were found in the morphology and structure of the larynx and pharynx. The morphology and density of bilateral parotid glands and submandibular glands are normal, and FDG physiological uptake occurs. The morphology and size of the thyroid gland were normal, with uniform density. No abnormalities were observed in FDG uptake. Several small lymph nodes were found in the deep cervical spaces and submandibular areas on both sides, and FDG uptake was increased in some lymph nodes. The translucency of both lungs increased, and cystic translucency shadows were seen under the pleura of the upper lobes of both lungs. The volume of the middle lobe of the right lung decreased, and patchy shadows of increased density were seen beside the hilum of the middle lobe of the right lung, with increased FDG uptake. There was no abnormality in the heart shadow. Partial arterial wall calcification (including coronary arteries). No esophageal dilation was observed, no obvious thickening or mass was found on the tube wall, and no increase in FDG uptake was observed. The liver margin was not smooth and neat, and the liver fissure did not show significant widening. A large, slightly low-density mass shadow was seen in the lower segment of the right lobe of the liver, with an indistinct boundary and increased FDG uptake. A slightly low-density nodule was seen in the left lobe of the liver, with a relatively clear boundary. FDG was uptake physiologically. A nodular dense shadow was also seen in the phrenic apex area of the right lobe of the liver. The main trunk and left and right branches of the portal vein were significantly thickened, with increased density and elevated FDG uptake. Lymph nodes in the portal space, hepatogastric space and retroperitoneum showed a mild increase in FDG uptake. The gallbladder was slightly plump, with no thickening of the gallbladder wall and no abnormalities in local FDG uptake. No abnormalities were observed in the morphology, size and density of the pancreas, and no significant abnormalities were found in FDG uptake. No abnormalities were observed in the morphology, size and density of the spleen, and no abnormalities were found in FDG uptake. Contrast agent residual images were observed in both urinary systems. The volume of the left kidney has shrunk. Multiple cystic low-density shadows and slightly high-density small nodules are seen in the left kidney, with clear boundaries and loss of FDG uptake. A fat-density nodule was observed at the upper pole of the right kidney, and no abnormalities were found in FDG uptake. No widening was observed in the bilateral renal pelvis, calyces and ureters. Bilateral adrenal imaging showed no obvious abnormalities, and FDG uptake was not abnormally increased. The stomach was moderately filled, with no significant thickening of the stomach wall. FDG uptake increased in some parts of the stomach wall. The intestinal filling was poor, and no obvious thickening or mass was observed in the intestinal wall. FDG was taken up physiologically. The prostate was slightly enlarged, and several calcification foci were seen in the parenchyma. There was no abnormal increase in FDG uptake. The bladder was not well filled, and a residual image of contrast agent was seen inside. Unstained reduction in bone density and osteophyte formation at the edges of some vertebrae in the spine. No abnormalities were observed in FDG uptake.
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Generated Report (MedDr)

Figure 7: Case Study: Examples of outputs following the template structure.

Case Study —— Output Following the Template Structure

Original Response (Chinese): 禁食状态下静脉注射18F-FDG，静息60分钟后行全身PET断层显像，全身断层影像显示：脑形态、结构正常，双侧大脑皮质放射性分布均匀、对称，FDG摄取未见明显异常。各脑室、脑沟、脑裂、脑池形态增宽，两侧脑室对称，中线结构无偏移。双侧眼球形态和轮廓无异常，FDG无异常摄取。右侧上颌窦黏膜稍增厚伴部分钙化，FDG摄取未见异常。鼻咽壁未见增厚，两侧腭扁桃体对称，FDG呈生理性摄取。喉咽形态、结构未见异常。双侧腮腺、颌下腺形态、密度正常，FDG生理性摄取。甲状腺形态、大小正常，密度均匀，FDG摄取未见异常。双侧颈深间隙、颌下见数枚小淋巴结显示，部分淋巴结FDG摄取增高。两肺透亮度增加，两肺上叶胸膜下见囊状透亮影。右肺中叶体积缩小，右肺中叶肺门旁见片状密度增高影，FDG摄取增高。心影未见异常。部分动脉管壁钙化（包括冠状动脉）。食管未见扩张，管壁未见明显增厚或肿块，FDG摄取未见增高。肝缘欠光整，肝裂未见明显增宽，CT平扫肝右叶下段见巨大稍低密度肿块影，边界欠清，FDG摄取增高。肝左叶见稍低密度结节，边界尚清晰，FDG摄取较基底摄取减低。肝右叶膈顶区另见结节状致密影。门静脉主干及左右支明显增粗，密度增高，FDG摄取增高。门腔间隙、肝胃间隙、腹膜后见淋巴结显示，FDG摄取轻度增高。胆囊稍饱满，胆囊壁未见增厚，局部FDG摄取未见异常。胰腺形态正常，胰体部见小结节状钙化灶，主胰管未见增宽，FDG摄取未见明显异常。脾脏形态、大小密度未见异常，FDG摄取未见异常。双侧泌尿系统见对比剂残留影；左肾体积缩小，左肾见多枚囊状低密度影及稍高密度小结节，边界清，FDG摄取缺失；右肾上极见一脂肪密度结节，FDG摄取未见异常。双侧肾盂肾盏及输尿管未见增宽。双侧肾上腺显影未见明显异常，FDG摄取未见异常。胃充盈尚可，胃壁未见明显增厚，部分胃壁FDG摄取增高。肠道充盈欠佳，肠壁未见明显增厚或肿块，FDG呈生理性摄取。前列腺稍增大，实质内见数枚钙化灶，FDG摄取未见异常增高。膀胱充盈欠佳，内见对比剂残留影。骨质密度不均匀减低，脊柱部分椎体边缘骨质增生。FDG摄取未见异常。 Translated English: Under fasting conditions, 18F-FDG was intravenously injected and then whole-body PET/CT tomography was performed after rest. After resting for 60 minutes, whole-body PET tomographic imaging was performed. The whole-body tomographic imaging showed that the brain morphology and structure were normal, the radioactive distribution of the bilateral cerebral cortex was uniform and symmetrical, and no obvious abnormalities were observed in FDG uptake. The morphology of each ventricle, sulcus, fissures and cisterns is widened, the bilateral ventricles are symmetrical, and the midline structure has no deviation. There were no abnormalities in the morphology and contour of both eyes, and no abnormal uptake of FDG. The mucosa of the right maxillary sinus was slightly thickened with partial calcification, and no abnormalities were observed in FDG uptake. The nasopharyngeal wall was not thickened, the palatine tonsils on both sides were symmetrical, and FDG was physiologically taken up. No abnormalities were found in the morphology and structure of the larynx and pharynx. The morphology and density of bilateral parotid glands and submandibular glands are normal, and FDG physiological uptake occurs. The morphology and size of the thyroid gland were normal, with uniform density. No abnormalities were observed in FDG uptake. Several small lymph nodes were found in the deep cervical Spaces and submandibular areas on both sides, and FDG uptake was increased in some lymph nodes. The translucency of both lungs increased, and cystic translucency shadows were seen under the pleura of the upper lobes of both lungs. The volume of the middle lobe of the right lung decreased, and patchy shadows of increased density were seen beside the hilum of the middle lobe of the right lung, with increased FDG uptake. There was no abnormality in the heart shadow. Partial arterial wall calcification (including coronary arteries). No esophageal dilation was observed, no obvious thickening or mass was found on the tube wall, and no increase in FDG uptake was observed. The liver margin was poorly trimmed, and no significant widening of the liver fissure was observed. On plain CT scan, a large, slightly low-density mass shadow was seen in the lower segment of the right lobe of the liver, with an indistinct boundary and increased FDG uptake. A slightly low-density nodule was observed in the left lobe of the liver, with a relatively clear boundary. The FDG uptake was lower than the background uptake. A nodular dense shadow was also seen in the phrenic apex area of the right lobe of the liver. The main trunk and left and right branches of the portal vein were significantly thickened, with increased density and elevated FDG uptake. Lymph nodes in the portal space, hepatogastric space and retroperitoneum showed a mild increase in FDG uptake. The gallbladder was slightly plump, with no thickening of the gallbladder wall and no abnormalities in local FDG uptake. The pancreatic calcification foci are seen in the pancreatic body. The main pancreatic duct is not widened, and no obvious abnormalities are observed in FDG uptake. No abnormalities were observed in the morphology, size and density of the spleen, and no abnormalities were found in FDG uptake. Contrast agent residual images were observed in both urinary systems. The volume of the left kidney has shrunk. Multiple cystic low-density shadows and slightly high-density small nodules are seen in the left kidney, with clear boundaries and loss of FDG uptake. A fat-density nodule was observed at the upper pole of the right kidney, and no abnormalities were found in FDG uptake. No widening was observed in the bilateral renal pelvis, calyces and ureters. No obvious abnormalities were found in bilateral adrenal imaging, and no abnormalities were observed in FDG uptake. The stomach was moderately filled, with no significant thickening of the stomach wall. FDG uptake increased in some parts of the stomach wall. The intestinal filling was poor, and no obvious thickening or mass was observed in the intestinal wall. FDG was taken up physiologically. The prostate was slightly enlarged, and several calcification foci were seen in the parenchyma. There was no abnormal increase in FDG uptake. The bladder was not well filled, and a residual image of contrast agent was seen inside. Uneven reduction in bone density and osteophyte formation at the edges of some vertebrae in the spine. No abnormalities were observed in FDG uptake.
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Generated Report (Lingshu-32B)

Original Response (Chinese): 禁食状态下静脉注射18F-FDG，静息60分钟后行全身PET断层显像，全身断层影像显示：- **脑部**：脑形态、结构正常，双侧大脑皮质放射性分布均匀、对称，FDG摄取未见明显异常。各脑室、脑沟、脑裂、脑池形态正常，两侧脑室对称，中线结构无偏移。- **头颈部**：双侧眼球形态和轮廓无异常，FDG无异常摄取。鼻咽壁未见增厚，两侧腭扁桃体对称，FDG呈生理性摄取。喉咽形态、结构未见异常。双侧腮腺、颌下腺形态、密度正常，FDG生理性摄取。甲状腺形态、大小正常，密度均匀，FDG摄取未见异常。双侧颈深间隙、颌下见数枚小淋巴结显示，部分淋巴结FDG摄取未见明显异常。- **胸部**：两肺透亮度正常，未见明显异常密度影，FDG摄取未见异常增高。心影未见异常，部分动脉管壁钙化（包括冠状动脉），FDG摄取未见异常增高。- **腹部**：食管未见扩张，管壁未见明显增厚或肿块，FDG摄取未见增高。肝缘光整，肝裂未见明显增宽，CT平扫未见明显异常密度影，FDG摄取未见异常增高。胆囊形态、大小正常，胆囊壁未见增厚，局部FDG摄取未见异常。胰腺形态正常，未见明显异常密度影，FDG摄取未见异常增高。脾脏形态、大小密度未见异常，FDG摄取未见异常增高。双侧泌尿系统未见明显异常密度影；双肾形态、大小、密度未见异常，FDG摄取未见异常增高。双侧肾盂肾盏及输尿管未见增宽，双侧肾上腺显影未见明显异常，FDG摄取未见异常增高。- **盆腔**：胃充盈尚可，胃壁未见明显增厚，FDG摄取未见异常增高。肠道充盈欠佳，肠壁未见明显增厚或肿块，FDG呈生理性摄取。前列腺形态、大小正常，实质内未见异常密度影，FDG摄取未见异常增高。膀胱充盈欠佳，内见对比剂残留影。- **骨骼**：骨质密度均匀，脊柱部分椎体边缘骨质增生。FDG摄取未见异常增高。综上所述，该患者全身FDG-PET/CT检查未见明显异常代谢增高灶，各脏器及骨骼系统未见异常密度影及代谢异常，未见肿瘤或其他病变征象。建议结合临床症状及其他检查结果进行综合评估。 Translated English: 18F-FDG was intravenously injected under fasting conditions. After 60 minutes of rest, whole-body PET tomography was performed. The whole-body tomography showed: - **Brain **: The brain morphology and structure were normal. The radioactive distribution of the bilateral cerebral cortex was uniform and symmetrical, and no obvious abnormalities were observed in FDG uptake. The morphology of each ventricle, sulcus, fissure and cisterns is normal. The bilateral ventricles are symmetrical, and there is no deviation in the midline structure. - ** Head and Neck **: There are no abnormalities in the shape and contour of both eyes, and no abnormal uptake of FDG. The nasopharyngeal wall was not thickened, the palatine tonsils on both sides were symmetrical, and FDG was physiologically taken up. No abnormalities were found in the morphology and structure of the larynx and pharynx. The morphology and density of bilateral parotid glands and submandibular glands are normal, and FDG physiological uptake occurs. The morphology and size of the thyroid gland were normal, with uniform density. No abnormalities were observed in FDG uptake. Several small lymph nodes were found in the deep cervical Spaces and submandibular areas on both sides, and no obvious abnormalities were observed in FDG uptake in some lymph nodes. - ** Chest **: The transparency of both lungs is normal, no obvious abnormal density shadows are observed, and FDG uptake is not abnormally increased. No abnormalities were found in the cardiac shadow. Some arterial walls calcified (including the coronary arteries), and no abnormal increase in FDG uptake was observed. - ** Abdomen **: No esophageal dilation was observed, no significant thickening or mass was found on the esophageal wall, and no increase in FDG uptake was observed. The liver margins were smooth, and no significant widening of the liver fissures was observed. No obvious abnormal density shadows were found on plain CT scan, and no abnormal increase in FDG uptake was observed. The morphology and size of the gallbladder are normal. There is no thickening of the gallbladder wall, and no abnormality in local FDG uptake is observed. The pancreatic morphology was normal, with no obvious abnormal density shadows observed, and no abnormal increase in FDG uptake was observed. There were no abnormalities in the morphology, size and density of the spleen, and no abnormal increase in FDG uptake was observed. No obvious abnormal density shadows were observed in the bilateral urinary systems. There were no abnormalities in the morphology, size and density of both kidneys, and no abnormal increase in FDG uptake was observed. No widening was observed in the bilateral renal pelvis, calyces and ureters. No obvious abnormalities were found in the bilateral adrenal imaging, and no abnormal increase in FDG uptake was observed. - ** Pelvic Cavity **: The stomach is moderately filled, with no significant thickening of the stomach wall and no abnormal increase in FDG uptake. The intestinal filling was poor, and no obvious thickening or mass was observed in the intestinal wall. FDG was taken up physiologically. The morphology and size of the prostate were normal. No abnormal density shadows were observed in the parenchyma, and no abnormal increase in FDG uptake was observed. The bladder was not well filled, and a residual image of contrast agent was seen inside. - ** Bones **: The bone density is uniform, and there is bone hyperplasia at the edges of some vertebrae in the spine. No abnormal increase in FDG uptake was observed. In conclusion, no obvious abnormal metabolic increase foci were found in the patient's whole-body FDG-PET/CT examination. No abnormal density shadows or metabolic abnormalities were observed in various organs and the skeletal system, and no signs of tumors or other lesions were found. It is recommended to conduct a comprehensive assessment in combination with clinical symptoms and other examination results.

Generated Report (Qwen2.5-VL-72B)

Figure 8: Case Study: Examples of outputs following the template structure.

Case Study ——— Output with Irrelevant Information

Original Response (Chinese): 好的，我将根据您提供的全身FDG-PET/CT图像，生成一份结构化的核医学报告。****核医学报告****检查项目：**** 全身18F-FDG PET/CT****检查目的：**** 评估全身18F-FDG代谢情况。查找潜在恶性肿瘤灶或其他异常。****检查方法：**** 禁食状态下静脉注射18F-FDG，静息60分钟后行全身PET断层显像，同步进行CT扫描。****报告日期：**** 2024年10月27日****报告医师：**** 经验丰富的核医学科医师****图像评估结果：**** ******1. 脑部：**双侧脑形态、结构正常，双侧大脑皮质放射性分布均匀、对称，FDG摄取未见明显异常。各脑室、脑沟、脑裂、脑池形态未见明显异常。****2. 头部软组织及颈部：**双侧眼球形态和轮廓无异常，FDG无异常摄取。右侧上颌窦黏膜未见明显增厚，FDG摄取未见异常。鼻咽壁未见增厚，两侧腭扁桃体对称，FDG呈生理性摄取。喉咽形态、结构未见异常。双侧腮腺、颌下腺形态、密度正常，FDG生理性摄取。甲状腺形态、大小正常，密度均匀，FDG摄取未见异常。双侧颈深间隙、颌下未见明显淋巴结显示，FDG无明显异常摄取。（*请注意：图像分辨率可能影响对小型淋巴结的评估。*）****3. 胸部：**双肺透亮度正常，未见明显囊状透亮影，未见明显体积缩小或片状密度增高影，FDG摄取未见明显异常。心影未见明显异常。部分动脉管壁未见明显钙化。纵隔未见明显淋巴结显示，FDG无明显异常摄取。****4. 腹部：**食管未见扩张，管壁未见明显增厚或肿块，FDG摄取未见明显增高。肝缘欠光整，肝裂未见明显增宽，FDG摄取未见明显异常。肝左叶未见明显低密度结节，FDG摄取未见异常。肝右叶未见明显结节状致密影，FDG摄取未见异常。门静脉主干及左右支未见明显增粗，密度未见明显增高，FDG摄取未见明显异常。门腔间隙、肝肾间隙、腹膜后未见明显淋巴结显示，FDG无明显异常摄取。胆囊未见明显异常，FDG摄取未见异常。胰腺形态正常，未见明显钙化灶或管壁增宽，FDG摄取未见明显异常。脾脏形态、大小密度未见异常，FDG摄取未见异常。****5. 泌尿系统：**双侧泌尿系统未见对比剂残留影；左肾形态、大小未见明显异常，未见多枚囊状低密度影或稍高密度小结节，FDG摄取未见明显异常；右肾未见明显脂肪密度结节，FDG摄取未见异常。双侧肾盂肾盏及输尿管未见增宽。双侧肾上腺显影未见明显异常，FDG摄取未见异常。****6. 消化系统：**胃充盈尚可，胃壁未见明显增厚，FDG摄取未见明显异常。肠道充盈欠佳，未见明显增厚或肿块，FDG呈生理性摄取。****7. 盆腔：**前列腺形态未见明显异常，未见明显钙化灶，FDG摄取未见异常增高。膀胱未见明显异常，内未见对比剂残留影。****8. 骨骼：**骨质密度未见明显均匀减低，脊柱未见明显椎体边缘骨质增生。FDG摄取未见明显异常。

Translated English: **Ok. I will generate a structured nuclear medicine report based on the whole-body FDG-PET/CT images you provided. ** Nuclear medicine report **** examination items: ** Whole body 18F-FDG PET/CT** Examination purpose: ** To assess the metabolism of 18F-FDG throughout the body and search for potential malignant lesions or other abnormalities. ** Examination method ** : 18F-FDG was intravenously injected in a fasting state. After resting for 60 minutes, whole-body PET tomography was performed, and CT scan was conducted simultaneously. ** Report Date: ** October 27, 2024 ** Reporting Physician: ** Experienced nuclear medicine physician ** Image Assessment result: ****1. Brain: The brain morphology and structure are normal. The radioactive distribution of the bilateral cerebral cortex is uniform and symmetrical. No obvious abnormalities were observed in FDG uptake. No obvious abnormalities were observed in the morphology of each ventricle, sulcus, fissure and cisterna. **2. Soft tissues of the head and neck: ** There are no abnormalities in the shape and contour of both eyes, and no abnormal uptake of FDG. No obvious thickening was observed in the mucosa of the right maxillary sinus, and no abnormalities were found in FDG uptake. The nasopharyngeal wall was not thickened, the palatine tonsils on both sides were symmetrical, and FDG was physiologically taken up. No abnormalities were found in the morphology and structure of the larynx and pharynx. The morphology and density of bilateral parotid glands and submandibular glands are normal, and FDG physiological uptake occurs. The morphology and size of the thyroid gland were normal, with uniform density. No abnormalities were observed in FDG uptake. No obvious lymph nodes were observed in the bilateral deep cervical space and submandibular space, and there was no obvious abnormal uptake of FDG. Please note: Image resolution may affect the assessment of small lymph nodes. *) **3. Chest: ** The translucency of both lungs is normal. No obvious cystic translucency shadows are observed. No significant volume reduction or patchy increase in density shadows are seen. No obvious abnormalities in FDG uptake are observed. No obvious abnormalities were found in the heart shadow. No obvious calcification was observed on the walls of some arteries. No obvious lymph nodes were observed in the mediastinum, and there was no significant abnormal uptake of FDG. **4. Abdomen: ** No esophageal dilation was observed, no significant thickening or mass was found on the tube wall, and no significant increase in FDG uptake was observed. The liver margin was underpolished, no significant widening of the liver fissures was observed, and no obvious abnormalities were found in FDG uptake. No obvious low-density nodules were observed in the left lobe of the liver, and no abnormalities were found in FDG uptake. No obvious nodular dense shadow was observed in the right lobe of the liver, and no abnormality was found in FDG uptake. The main portal vein and its left and right branches showed no significant thickening, no significant increase in density, and no obvious abnormalities in FDG uptake. No obvious lymph nodes were found in the portal space, hepatogastric space or retroperitoneum, and there was no obvious abnormal uptake of FDG. No obvious abnormalities were found in the gallbladder, and no abnormalities were observed in FDG uptake. The pancreatic morphology was normal, with no obvious calcification foci or widened duct walls, and no significant abnormalities in FDG uptake were observed. No abnormalities were observed in the morphology, size and density of the spleen, and no abnormalities were found in FDG uptake. **5. Urinary system: ** No contrast agent residual images were observed in both urinary systems. There were no obvious abnormalities in the morphology and size of the left kidney. No multiple cystic low-density shadows or slightly high-density small nodules were observed, and no obvious abnormalities were found in FDG uptake. No obvious fat density nodules were observed in the right kidney, and no abnormalities were found in FDG uptake. No widening was observed in the bilateral renal pelvis, calyces and ureters. No obvious abnormalities were found in bilateral adrenal imaging, and no abnormalities were observed in FDG uptake. 6. Digestive System: The stomach is moderately filled, with no significant thickening of the stomach wall and no obvious abnormalities in FDG uptake. The intestinal filling was poor, with no obvious thickening or mass observed. FDG was taken up physiologically. 7. Pelvic cavity: No obvious abnormalities were observed in the morphology of the prostate, no obvious calcification foci were found, and no abnormal increase in FDG uptake was observed. No obvious abnormalities were observed in the bladder, and no residual contrast agent was found inside. **8. Bones: ** No significant uniform reduction in bone density was observed, and no obvious osteophytes were found at the edges of the vertebral bodies in the spine. No obvious abnormalities were observed in FDG uptake.**

Generated Report (MedGemma-27B)

Original Response (Chinese): 患者信息姓名：张三性别：男年龄：45岁临床信息1.病史：无特殊病史。2.体格检查：无明显异常。3.既往史：无特殊病史。4.家族史：无特殊病史。影像学检查1. PET/CT影像：患者进行禁食状态静脉注射18F-FDG，静息60分钟后行全身PET断层显像。全身断层影像显示：脑形态、结构正常，双侧大脑皮质放射性分布均匀、对称，FDG摄取未见明显异常。各脑室、脑沟、脑裂、脑池形态正常，中线结构无偏移。2. PET异常区域：双侧眼球形态和轮廓无异常，FDG无异常摄取。右侧上颌窦黏膜稍增厚伴部分钙化，FDG摄取未见异常。鼻咽壁未见增厚，两侧腭扁桃体对称，FDG呈生理性摄取。喉咽形态、结构未见异常。双侧腮腺、颌下腺形态、密度正常，FDG生理性摄取。甲状腺形态、大小正常，密度均匀，FDG摄取未见异常。双侧颈深间隙、颌下见数枚小淋巴结显示，部分淋巴结FDG摄取增高。3. 两肺透亮度增加，两肺上叶胸膜下见囊状透亮影。右肺中叶体积缩小，右肺中叶肺门旁见片状密度增高影，FDG摄取增高。心影未见异常。部分动脉管壁钙化（包括冠状动脉）。4. 食管未见扩张，管壁未见明显增厚或肿块，FDG摄取未见增高。肝缘欠光整，肝裂未见明显增宽，CT平扫肝右叶下段见巨大稍低密度肿块影，边界欠清，FDG摄取增高。肝左叶见稍低密度结节，边界尚清晰，FDG摄取较本底摄取减低。肝右叶膈顶区另见结节状致密影。门静脉主干及左右支明显增粗，密度增高，FDG摄取增高。门腔间隙、肝肾间隙、腹膜后见淋巴结显示，FDG摄取程度增高。5. 胆囊稍饱满，胆囊壁未见增厚，局部FDG摄取未见异常。胰腺形态正常，胰体部见小结节状钙化灶，主胰管未见增宽，FDG摄取未见明显异常。脾脏形态、大小密度未见异常，FDG摄取未见异常。双侧泌尿系统未见对比剂残留影；左肾体积缩小，左肾见多枚囊状低密度影及稍高密度小结节，边界清，FDG摄取缺失；右肾上极见一脂肪密度结节，FDG摄取未见异常。双侧肾盂肾盏及输尿管未见增宽。双侧肾上腺显影未见明显异常，FDG摄取未见异常。6. 胃充盈尚可，胃壁未见明显增厚，部分胃壁FDG摄取增高。肠道充盈欠佳，肠壁未见明显增厚或肿块，FDG呈生理性摄取。前列腺稍增大，实质内见数枚钙化灶，FDG摄取未见异常增高。膀胱充盈欠佳，内见对比剂残留影。7. 骨质密度不均匀减低，脊柱部分椎体边缘骨质增生。FDG摄取未见异常。

Translated English: **Patient Information:** Name: Zhang SAN Gender: Male Age: 45 Years Old **Clinical Information 1. Medical History:** No special medical history. **2. Physical examination:** No obvious abnormalities. **3. Past medical history:** No special medical history. **4. Family history:** No special medical history. **Imaging examinations:** 1. PET/CT images: The patient was intravenously injected with 18F-FDG while fasting. After 60 minutes of rest, a whole-body PET tomographic image was performed. The whole-body tomographic image showed that the brain morphology and structure were normal, the radioactive distribution of the bilateral cerebral cortex was uniform and symmetrical, and no obvious abnormalities were found in FDG uptake. The morphology of each ventricle, sulcus, fissures and cisterns is widened, the bilateral ventricles are symmetrical, and the midline structure has no deviation. 2. Abnormal PET areas: There are no abnormalities in the shape and contour of both eyes, and no abnormal FDG uptake. The mucosa of the right maxillary sinus was slightly thickened with partial calcification, and no abnormalities were observed in FDG uptake. The nasopharyngeal wall was not thickened, the palatine tonsils on both sides were symmetrical, and FDG was physiologically taken up. No abnormalities were found in the morphology and structure of the larynx and pharynx. The morphology and density of bilateral parotid glands and submandibular glands are normal, and FDG physiological uptake occurs. The morphology and size of the thyroid gland were normal, with uniform density. No abnormalities were observed in FDG uptake. Several small lymph nodes were found in the deep cervical Spaces and submandibular areas on both sides, and FDG uptake was increased in some lymph nodes. 3. The translucency of both lungs increased, and cystic translucency shadows were seen under the pleura of the upper lobes of both lungs. The volume of the middle lobe of the right lung decreased, and patchy shadows of increased density were seen beside the hilum of the middle lobe of the right lung, with increased FDG uptake. There was no abnormality in the heart shadow. Partial arterial wall calcification (including coronary arteries). 4. No esophageal dilation was observed, no significant thickening or mass was found on the tube wall, and no increase in FDG uptake was observed. The liver margin was poorly trimmed, and no significant widening of the liver fissure was observed. On plain CT scan, a large, slightly low-density mass shadow was seen in the lower segment of the right lobe of the liver, with an indistinct boundary and increased FDG uptake. A slightly low-density nodule was observed in the left lobe of the liver, with a relatively clear boundary. The FDG uptake was lower than the background uptake. A nodular dense shadow was also seen in the phrenic apex area of the right lobe of the liver. The main trunk and left and right branches of the portal vein were significantly thickened, with increased density and elevated FDG uptake. Lymph nodes in the portal space, hepatogastric space and retroperitoneum showed a mild increase in FDG uptake. 5. The gallbladder was slightly plump, with no thickening of the gallbladder wall and no abnormalities in local FDG uptake. The pancreatic morphology is normal. Small nodular calcification foci are seen in the pancreatic body. The main pancreatic duct is not widened, and no obvious abnormalities are observed in FDG uptake. No abnormalities were observed in the morphology, size and density of the spleen, and no abnormalities were found in FDG uptake. Contrast agent residual images were observed in both urinary systems. The volume of the left kidney has shrunk. Multiple cystic low-density shadows and slightly high-density small nodules are seen in the left kidney, with clear boundaries and loss of FDG uptake. A fat-density nodule was observed at the upper pole of the right kidney, and no abnormalities were found in FDG uptake. No widening was observed in the bilateral renal pelvis, calyces and ureters. No obvious abnormalities were found in bilateral adrenal imaging, and no abnormalities were observed in FDG uptake. 6. The stomach was moderately filled, with no significant thickening of the stomach wall. FDG uptake increased in some parts of the stomach wall. The intestinal filling was poor, and no obvious thickening or mass was observed in the intestinal wall. FDG was taken up physiologically. The prostate was slightly enlarged, and several calcification foci were seen in the parenchyma. There was no abnormal increase in FDG uptake. The bladder was not well filled, and a residual image of contrast agent was seen inside. 7. Uneven reduction in bone density and osteophyte formation at the edges of some vertebrae in the spine. No abnormalities were observed in FDG uptake.

Generated Report (MedVLM-R1)

Figure 9: Case Study: Examples of irrelevant information not involved in the report template.

Case Study — Unstructured Outputs

Original Response (Chinese): 在PET/CT图像中，我们可以观察到双肺的透亮度增加和囊状透明影。右肺中叶体积缩小并存在片状密度增高影，而左肺则有结节状致密影。脾脏形态、大小和密度未见异常，但我们可以看到部分胸腺钙化（包括冠状动脉）。肝缘欠光整，肝裂未见明显增宽，CT平扫肝右叶下段有一个巨大稍低密度肿块影，边界欠清。门静脉主干及左右支明显增粗，密度增高。胆囊稍饱满，胆囊壁未见增厚。左肾体积缩小，左肾有多枚囊状低密度影及稍高密度小结节，边界清。双侧泌尿系统未见明显异常。腹膜后可以看到淋巴结显示，FDG摄取缺失。双侧肾上腺显影未见明显异常。胃充盈尚可，胃壁未见明显增厚或肿块。肠道充盈欠佳，肠壁未见明显增厚或肿块。前列腺稍增大，内有数枚钙化灶，FDG摄取未见异常增高。膀胱充盈欠佳，内见对比剂残留影。骨质密度不均匀减低，脊柱部分椎体边缘骨质增生。FDG无异常摄取。

Translated English: In PET/CT images, we can observe an increase in the translucency of both lungs and cystic hyaline shadows. The volume of the middle lobe of the right lung has shrunk and there are patchy shadows of increased density, while there are nodular dense shadows in the left lung. The morphology, size and density of the spleen showed no abnormalities, but we could observe partial pleural calcification (including coronary arteries). The liver margin was poorly trimmed, and no significant widening of the liver fissure was observed. On a plain CT scan of the lower segment of the right lobe of the liver, there was a large, slightly low-density mass shadow with an indistinct boundary. The main trunk and left and right branches of the portal vein have significantly thickened and increased density. The gallbladder is slightly plump, and the gallbladder wall shows no thickening. The volume of the left kidney has shrunk. There are multiple cystic low-density shadows and slightly high-density small nodules in the left kidney, with clear boundaries. No obvious abnormalities were found in the bilateral urinary systems. Lymph nodes can be seen in the retroperitoneum, and FDG uptake is absent. Bilateral adrenal imaging showed no obvious abnormalities. The stomach was moderately filled, and no obvious thickening or mass was observed on the stomach wall. The intestinal filling was poor, and no obvious thickening or mass was observed in the intestinal wall. The prostate was slightly enlarged, with several calcification foci inside, and FDG uptake was not abnormally elevated. The bladder was not well filled, and a residual image of contrast agent was seen inside. Uneven reduction in bone density and osteophyte formation at the edges of some vertebrae in the spine. There was no abnormal uptake of FDG.

Generated Report (OmniLMM)

Original Response (Chinese): 形态、大小正常，FDG摄取未见异常。胰腺头部密度均匀，边界清楚，FDG生理性摄取。脾脏形态、大小正常，FDG摄取较本底摄取减低。双侧肾上腺形态、大小正常，FDG无明显异常摄取。左肾孟区可见轻度扩张影，右输尿管中段局部狭窄伴上游积水征象，右肾实质内见斑片状稍高密度灶影，提示可能的结石或肿块存在。膀胱壁增厚并呈不规则形变，右侧腹股沟区见小淋巴结显示，部分淋巴结FDG摄取增高。根据提供的PET/CT影像，请撰写核医学报告：禁食状态下静脉注射18F-FDG，静息60分钟后行全身PET断层显像，全身断层影像显示：脑形态、结构正常，双侧大脑皮质放射性分布均匀，对称，FDG摄取未见明显异常。各脑室、脑沟、脑裂、脑池形态增宽，两侧脑室对称，中线结构无偏移。双侧眼球形态和轮廓无异常，FDG无异常摄取。右侧上颌窦黏膜稍增厚伴部分钙化，FDG摄取未见异常。鼻咽壁未见增厚，两侧腭扁桃体对称，FDG呈生理性摄取。喉咽形态、结构未见异常。双侧腮腺、颌下腺形态、密度正常，FDG生理性摄取。甲状腺形态、大小正常，密度均匀，FDG摄取未见异常。双侧颈深沟隙、颌下见数枚小淋巴结显示，部分淋巴结FDG摄取增高两肺透亮度增加，两肺上叶胸膜下见囊状透亮影。右肺中叶体积缩小，右肺中叶肺门旁见片状密度增高影，FDG摄取增高。心影未见异常。部分动脉管壁钙化（包括冠状动脉）。食管未见扩张，管壁未见明显增厚或肿块，FDG摄取未见增高。肝缘欠光整，肝裂未见明显增宽，CT平扫肝右叶下段见巨大稍低密度肿块影，边界欠清，FDG摄取增高。肝左叶见稍低密度结节，边界尚清晰，FDG摄取较本底摄取减低。肝右叶膈顶区另见结节状致密影。门静脉主干及左右支明显增粗，密度增高，FDG摄取增高。门腔间隙、肝胃间隙、腹膜后见淋巴结显示，FDG摄取轻度增高。胆囊形态、大小正常，FDG摄取未见异常。胰腺头部密度均匀，边界清楚，FDG生理性摄取。脾脏形态、大小正常，FDG摄取较本底摄取减低。双侧肾上腺形态、大小正常，FDG无明显异常摄取。左肾孟区可见轻度扩张影，右输尿管中段局部狭窄伴上游积水征象，右肾实质内见斑片状稍高密度灶影，提示可能的结石或肿块存在。膀胱壁增厚并呈不规则形变，右侧腹股沟区见小淋巴结显示，部分淋巴结FDG摄取增高。

Translated English: The shape and size are normal, and no abnormalities were found in FDG uptake. The head and body of the pancreas have uniform density, clear boundaries, and FDG physiological uptake. The morphology and size of the spleen are normal, and the FDG uptake is lower than the background uptake. The morphology and size of both adrenal glands are normal, and there is no obvious abnormal uptake of FDG. A mild dilation shadow can be seen in the left renal pelvis area. There is a local stenosis in the middle section of the right ureter accompanied by effusion upstream. In the right renal parenchyma, there are patchy slightly high-density foci, suggesting the possible presence of stones or masses. The bladder wall is thickened and shows irregular deformation. Small lymph nodes are seen in the right groin area, and FDG uptake is increased in some lymph nodes. Based on the provided PET/CT images, please write the nuclear medicine report: 18F-FDG was intravenously injected in a fasting state. After 60 minutes of rest, a whole-body PET tomographic image was performed. The whole-body tomographic image showed that the brain morphology and structure were normal, the radioactive distribution of the bilateral cerebral cortex was uniform and symmetrical, and no obvious abnormalities were found in FDG uptake. The morphology of each ventricle, sulcus, fissures and cisterns is widened, the bilateral ventricles are symmetrical, and the midline structure has no deviation. There were no abnormalities in the morphology and contour of both eyes, and no abnormal uptake of FDG. The mucosa of the right maxillary sinus was slightly thickened with partial calcification, and no abnormalities were observed in FDG uptake. The nasopharyngeal wall was not thickened, the palatine tonsils on both sides were symmetrical, and FDG was physiologically taken up. No abnormalities were found in the morphology and structure of the larynx and pharynx. The morphology and density of bilateral parotid glands and submandibular glands are normal, and FDG physiological uptake occurs. The morphology and size of the thyroid gland were normal, with uniform density. No abnormalities were observed in FDG uptake. Several small lymph nodes were seen in the deep cervical Spaces and submandibular areas on both sides, with increased FDG uptake in some lymph nodes. The translucency of both lungs increased, and cystic translucency shadows were observed under the pleura of the upper lobes of both lungs. The volume of the middle lobe of the right lung decreased, and patchy shadows of increased density were seen beside the hilum of the middle lobe of the right lung, with increased FDG uptake. A nodular dense shadow was also seen in the phrenic apex area of the right lobe of the liver. No esophageal dilation was observed, no obvious thickening or mass was found on the tube wall, and no increase in FDG uptake was observed. The liver margin was poorly trimmed, and no significant widening of the liver fissure was observed. On plain CT scan, a large, slightly low-density mass shadow was seen in the lower segment of the right lobe of the liver, with an indistinct boundary and increased FDG uptake. A slightly low-density nodule was observed in the left lobe of the liver, with a relatively clear boundary. The FDG uptake was lower than the background uptake. A nodular dense shadow was also seen in the phrenic apex area of the right lobe of the liver. The main trunk and left and right branches of the portal vein were significantly thickened, with increased density and elevated FDG uptake. Lymph nodes in the portal space, hepatogastric space and retroperitoneum showed a mild increase in FDG uptake. The morphology and size of the gallbladder are normal, and no abnormalities were observed in FDG uptake. The head and body of the pancreas have uniform density, clear boundaries, and FDG physiological uptake. The morphology and size of the spleen are normal, and the FDG uptake is lower than the background uptake. The morphology and size of both adrenal glands are normal, and there is no obvious abnormal uptake of FDG. A mild dilation shadow can be seen in the left renal pelvis area. There is a local stenosis in the middle section of the right ureter accompanied by effusion upstream. In the right renal parenchyma, there are patchy slightly high-density foci, suggesting the possible presence of stones or masses. The bladder wall is thickened and shows irregular deformation. Small lymph nodes are seen in the right groin area, and FDG uptake is increased in some lymph nodes.

Generated Report (HuatuoGPT-Vision)

Original Response (Chinese): **核医学报告****患者信息：**患者姓名: 年龄, 性别**检查日期**检查目的：**评估全身代谢活动及异常病灶**检查方法：**静脉注射18F-FDG，静息60分钟后行全身PET断层显像**主要发现：**1.**肺部：** - 形态、结构正常 - 双侧大脑皮质放射性分布均匀、对称 - FDG摄取未见明显异常2.**眼部：** - 形态和轮廓无异常 - FDG无异常摄取3.**鼻咽部：** - 鼻咽壁未见增厚，两侧腭扁桃体对称 - FDG呈生理性摄取4.**喉咽部：** - 形态、结构未见异常5.**唾液腺：** - 形态、密度正常 - FDG生理性摄取6.**甲状腺：** - 形态、大小正常，密度均匀 - FDG摄取未见异常7.**颈部淋巴结：** - 见数枚小淋巴结显示，部分淋巴结FDG摄取增高8.**肺部：** - 右肺中叶体积缩小，右肺中叶肺门旁见片状密度增高影 - FDG摄取增高9.**心脏：** - 未见异常，部分动脉管壁钙化（包括冠状动脉）10.**食管：** - 未见扩张，管壁未见明显增厚或肿块 - FDG摄取未见增高11.**肝脏：** - 肝右叶下段见巨大稍低密度肿块影，边界欠清 - FDG摄取增高 - 肝左叶见稍低密度结节，边界尚清晰 - FDG摄取较本底摄取减低 - 肝右叶膈顶区另见结节状致密影 - FDG摄取增高 - 门静脉主干及左右支明显增粗，密度增高 - FDG摄取增高 - 门腔间隙、肝胃间隙、腹膜后见淋巴结显示 - FDG摄取轻度增高12.**胆囊：** - 稍饱满，胆囊壁未见增厚 - 局部FDG摄取未见异常13.**胰腺：** - 形态正常，胰体部见小结节状钙化灶，主胰管未见增宽 - FDG摄取未见明显异常14.**脾脏：** - 形态、大小密度未见异常 - FDG摄取未见异常15.**泌尿系统：** - 左肾体积缩小，左肾见多枚囊状低密度影及稍高密度小结节，边界清 - FDG摄取缺失 - 右肾上腺一脂肪密度结节 - FDG摄取未见异常16.**肾脏及肾上腺：** - 肾盂肾盏及输尿管未见增宽 - 双侧肾上腺显影未见明显异常 - FDG摄取未见异常17.**胃部：** - 充盈尚可，胃壁未见明显增厚 - 部分胃壁FDG摄取增高18.**肠道：** - 充盈欠佳，肠壁未见明显增厚或肿块 - FDG呈生理性摄取19.**前列腺：** - 稍增大，实质内见数枚钙化灶 - FDG摄取未见异常增高20.**膀胱：** - 充盈欠佳，内见对比剂残留影21.**骨骼系统：** - 骨质密度不均匀减低 - 脊柱部分椎体边缘骨质增生 - FDG摄取未见异常**结论：** PET图像显示多个代谢增高病灶，包括肝脏、肺、门静脉、胰腺及骨骼系统。CT图像显示肝脏巨大肿块影，右肺中叶体积缩小，左肾体积缩小并见多枚囊状低密度影及稍高密度小结节。建议进一步明确病灶性质，建议结合临床及影像学资料进行综合评估。

Translated English: In PET/CT images, we can observe an increase in the translucency of both lungs and cystic hyaline shadows. The volume of the middle lobe of the right lung has shrunk and there are patchy shadows of increased density, while there are nodular dense shadows in the left lung. The morphology, size and density of the spleen showed no abnormalities, but we could observe partial pleural calcification (including coronary arteries). The liver margin was poorly trimmed, and no significant widening of the liver fissure was observed. On a plain CT scan of the lower segment of the right lobe of the liver, there was a large, slightly low-density mass shadow with an indistinct boundary. The main trunk and left and right branches of the portal vein have significantly thickened and increased density. The gallbladder is slightly plump, and the gallbladder wall shows no thickening. The volume of the left kidney has shrunk. There are multiple cystic low-density shadows and slightly high-density small nodules in the left kidney, with clear boundaries. No obvious abnormalities were found in the bilateral urinary systems. Lymph nodes can be seen in the retroperitoneum, and FDG uptake is absent. Bilateral adrenal imaging showed no obvious abnormalities. The stomach was moderately filled, and no obvious thickening or mass was observed on the stomach wall. The intestinal filling was poor, and no obvious thickening or mass was observed in the intestinal wall. The prostate was slightly enlarged, with several calcification foci inside, and FDG uptake was not abnormally elevated. The bladder was not well filled, and a residual image of contrast agent was seen inside. Uneven reduction in bone density and osteophyte formation at the edges of some vertebrae in the spine. There was no abnormal uptake of FDG.

Generated Report (DeepSeek-VL2)

Figure 10: Case Study: Examples of unstructured outputs without following the report template.

Case Study — Refuse to Answer

Original Response: Error code: 400 - 'error': {'code': 'data_inspection_failed', 'param': None, 'message': 'Input data may contain inappropriate content.', 'type': 'data_inspection_failed', 'id': 'chatcmpl-33dcad52-cd3b-9b21-a054-42233500542a', 'request_id': '33dcad52-cd3b-9b21-a054-42233500542a'}

Generated Report (Qwen-VL-max)

Case Study — Chaotic / Empty Outputs

[illegible]

Generated Report (RadFM)

Original Response (Chinese): 全身PET/CT检查显示，SUV值为2.5，且存在明显的肺部高分辨率影，这可能是由肺部感染引起的。此外，该区域内还存在肝脏结节以及肾脏的脂肪密度结。需要进一步进行CT扫描或其他影像学检查来确定具体病因。建议患者在医生的指导下定期进行随访和治疗，以最大程度地改善生活质量和健康结局。

Translated English: whole-body PET/CT examination showed that the SUV value was 2.5, and there was a distinct high-resolution image of the lungs, which might be caused by a lung infection. In addition, there are liver nodules and fat density nodules in the kidneys in this area. Further CT scans or other imaging examinations are needed to determine the specific cause. It is recommended that patients undergo regular follow-ups and treatments under the guidance of doctors to maximize the improvement of their quality of life and health outcomes.

Generated Report (VisualGLM)

Original Response (Chinese): 镶嵌针孔
Translated English: Inlaid pinholes

Generated Report (Qilin-Med-VL)

Original Response (Chinese): 核医学报告

Translated English: Nuclear Medicine Report

Generated Report (Yi-VL-6B)

Original Response (Chinese): ""
Translated English: ""

Generated Report (Med-Flamingo)

Figure 11: Case Study: Examples of refuse to answer and chaotic/empty outputs.